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Nexora - Redefining Interview Readiness

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Final Year Project

November 24, 2025

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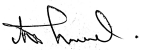
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Abstract

Nexora is an interview preparation tool which helps the candidate to practice and improve their domain-specific interview preparation by analysing verbal and non-verbal communication based on the input provided through the microphone and web camera. The system will support multilingual speech-to-text in English, Urdu, French, and Spanish and the system will know when a candidate has finished an answer either by inactivity in speech-to-text or by clicking on the next question. Each reply is evaluated by means of semantic large-language model, and non-verbal cues are tracked with a visual indicator that displays green as the confirmation of the proper behaviour and red as a sign of distracting actions. At the end of the practice session, Nexora will produce a detailed report that includes question-specific scores, overall score, and individualised improvement recommendations. It is a multimodal method that offers a more efficient, organized, and accessible experience of preparing interviews based on specific areas of professionalism.

Executive Summary

Recruitment is an important process for future employees and organizations, yet it is often devoid of a system that ensures fast feedback for interviews. Candidates frequently do not know how they have performed until long after the interview, which limits their ability to make timely improvements. Nexora is a domain specific AI-powered interview preparation platform that addresses this gap by providing real-time feedback and multimodal analysis, refining conventional approaches to interview practice.

With input from both microphone and webcam, Nexora evaluates verbal and non-verbal communication of candidates, making interviews more interactive and effective. Multilingual speech-to-text transcription in English, Urdu, French, and Spanish enables a diverse range of candidates to participate seamlessly. The system intelligently detects when a candidate has finished their response either through a short pause or by clicking the “Next Question” button before sending the complete answer to a semantic large language model (LLM). The LLM provides detailed per-question scores, highlights strengths and weaknesses, and delivers personalized improvement recommendations, ensuring context-aware assessment without relying on keyword-based evaluation and minimizing unnecessary LLM usage.

Nexora also conducts real-time non-verbal analysis using webcam input. It monitors facial expressions and gestures, such as unnecessary hand movements, which may distract interviewers. Real-time feedback is provided through a dynamic visual indicator: green signals appropriate behavior, while red signals distracting actions. This allows candidates to adjust their body language during the interview, improving overall presentation.

All assessments including transcripts, question-wise scores, behavioral observations, and AI feedback are stored in a centralized database. They are compiled into detailed performance report that show overall scores, per-question evaluations, non-verbal behavior insights, identified strengths and weaknesses, and personalized recommendations. This comprehensive reporting helps candidates understand their readiness for real interviews.

A systematic review of current interview-preparation systems reveals that most overlook non-verbal cues, provide feedback only after the session, or support only English-speaking users. Nexora overcomes these limitations by offering a real-time, multilingual, and holistic solution. By bridging the gap between traditional interviews and modern AI-driven assessments, Nexora helps candidates improve their communication skills, boost their confidence, and better prepare for real interviews. Through live interaction, multimodal AI-powered scoring, and comprehensive performance reporting, the platform offers a structured, data-driven, and engaging approach to interview preparation, setting a new standard in effectiveness, inclusivity, and impact for candidates worldwide.

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Chapter 1

Introduction

The traditional job interview process has been traditionally based on face-to-face communication and personal decisions made by interviewers. This results in the situation where the candidates are often unaware of their performance until the interview process is over, a situation that significantly hinders real-time improvement. Even though digital tools and training platforms have tried to mitigate this deficiency, they typically provide feedback at the end of a session with little consideration of non-verbal signals, which are often important, such as body language, gestures, and facial expressions. In addition, most of the solutions that have survived are monolingual, thus limiting accessibility by those who do not speak English. To fill these gaps, the current project, Nexora, aims at changing the preparation stage of interview engagement. Nexora, through the provision of a web-based platform coupled with real-time audio-video analytics, will be aimed at providing the candidate with immediate visual information and assessive feedback during the interview process. Besides real-time audio and video analysis, Nexora is an evaluator that uses a semantic LLM to assess the content of the answers given by the candidate. The system identifies when a candidate has completed an answer, either by a brief pause in speech or by clicking the button of the next question, thus allowing immediate context-informed feedback as to the quality and relevance of their answers. The monitoring application of the system evaluates verbal and non-verbal communication, making distracting behaviour a red signal and proper behaviour a green signal. Moreover, every answer or response is subjected to semantic analysis to determine its clarity, relevancy, and domain specificity thus coming up with recommendation to take some action on its delivery and content. Furthermore, Nexora supports various languages, such as English, Urdu, French, and Spanish, hence ensuring inclusivity to a diverse world population. The methodology is supposed to hasten the preparation process, increase confidence of the candidates, create transparency and fairness in the process of evaluating performance during interviews.

1.1 Purpose of this Document

This report is a technical and conceptual report about our Final Year Project, Nexora . It describes the motivation, the goals of the project, the design, the methodology, implementation and evaluation. The main aim of the project is to build a web- based platform that helps candidates by providing real-time feedback of interviews. This system allows candidates to correct their communication immediately, rather than waiting until after the fact. It also offers context-sensitive analysis of verbal answers, which helps the applicants to establish their strengths, weaknesses, and improvements in every response they give.

The project aims to provide a Minimum Viable Product (MVP) that will include:

- Audio and video real-time monitoring.
- A visual-based AI-powered feedback.
- Semantic analysis of content presented.
- English, Urdu, French, and Spanish multilingualism.

The interface has been created to be user-friendly and improve user experience and accessibility. The entire procedure of designing, developing, testing and evaluation of the platform is recorded in this report and the limitations of the platform and future improvements are discussed.

1.2 Intended Audience

This report is targeted at the following audience:

- The evaluation team consisting of academic appraisers, including supervisors and a team of professors, who will review the design, implementation, and contribution of the project.
- Candidates preparing for interviews can use this system to practice and improve their communication skills, making it a valuable tool for professional development.
- Students and researchers in Computer Science, Artificial Intelligence, and Human-Computer Interaction may find this project a reference within their future research or project.

To a general, non-technical user, who might be using Nexora in future, a simplified functionality and usage of the system will be made available in user documentation or training content.

1.3 Definitions, Acronyms, and Abbreviations

Important definitions, acronyms, and abbreviations used in this document:

SDG: Sustainable Development Goal

FYP: Final Year Project

MVP: Minimum Viable Product

UI: User Interface

UX: User Experience

AI: Artificial Intelligence

Agile: Agile Development

SCRUM: Scrum Development

REST: Representational State Transfer

HCI: Human–Computer Interaction

NLP: Natural Language Processing

ORM: Object-Relational Mapping

CRUD: Create, Read, Update, Delete

API: Application Programming Interface

LLM: Large Language Model

1.4 Conclusion

Finally, the conventional job interviewing has the disadvantage of providing timely and substantive feedback to the candidates, thus inhibiting their ability to understand areas that require improvement during the assessment procedure. Even though some of the digital solutions exist, they mainly focus on verbal aspects and often overlook some important non-verbal signals, such as gestures, tone, eye engagement, and facial expression. In addition, these solutions normally release feedback on completion of the interview session, thus reducing their effectiveness in real-time improvement. Nexora tackles this shortcoming by designing a mechanism that allows for audio and video analysis, thus giving instant feedback while being interviewed. It also scores response content with a semantic LLM so that candidates can also get robust feedback on not only delivery but also the content of their answers, ultimately leading to better interview preparedness. The platform is designed to offer an easier, more open, and efficient interview preparation experience through real-time tracking, immediate visual cues, and multilingual support. This report provides a comprehensive overview of Nexora, covering its background, main goals, design, implementation, assessment, and future developments, and highlights the contributions it makes to improving interview preparation.

Chapter 2

Project Vision

This chapter provides the overall vision and roadmap of the Nexora initiative. It demarcates the domain of the problem, discusses the existing predicaments of the interview preparation, and emphasizes the role of Nexora in providing AI-based solutions to those problems. The chapter also outlines the project's goals, objectives, and scope, providing a clear understanding of the system's intended purpose and impact.

2.1 Problem Domain Overview

The current research is dedicated to the creation of an AI-powered platform that will be used to facilitate job-interview preparation. Traditional interview systems are based on human rating and usually provide feedback at the end of the session, thus denying the applicant an opportunity to determine the mistakes committed during the interview. Conversely, the Nexora system automatically conducts interviews, real-time performance appraisals and provides immediate instructional feedback.

First time users must be registered on the site. After successful registration, the candidates can start a simulated interview where the AI interviewer asks questions and analyzes the responses of the applicants. Sound and video messages are received through microphone and web camera respectively. These signals are then processed using advanced machine-learning models which evaluate non-verbal cues, such as facial expressions, gestures, tone, and eye engagement etc. While non-verbal signals are checked throughout and processed in real time, the content of each candidate's response is captured by the LLM and scored only when that response is finished. This process ensures accurate scoring and feedback that is then used to generate the performance report in the end.

Real-time feedback is provided through visual indicators:

- A green icon indicates that the actions of the applicant and communicative performance are of the required standards.
- A red icon is a sign of distractions, such as overuse of gestures, or unstable eye contact.

In addition to the delivery of feedback, Nexora provides multilingual assistance in English, Urdu, French, and Spanish and thus fosters inclusiveness among applicants in the multilingual environment. This polyglot approach will increase the flexibility of the platform to a global user base, which will increase its usefulness in a heterogeneous context.

The overall mission of Nexora is to provide the candidates with a self-directed AI-enhanced setting that supplements the interview readiness. The combination of audio-visual surveillance and instant visual feedback makes the platform allow refining the communicative skills, strengthening self-confidence, and gaining experience, which is independent of human reviewers. As a result, the process achieves an increased interactivity, transparency, and accessibility worldwide.

2.2 Problem Statement

The existing systems and practices that are committed to interview preparation have significant weaknesses especially in the fact that they do not offer real time feedback to applicants. Most of the systems that are in place focus on verbal communication only, provide post-session assessment, or are not very inclusive due to the limitations of monolingualism. To address these shortcomings, we are creating Nexora, a platform based on artificial intelligence that conducts interviews and provides real-time, two-channel (verbal and non-verbal) feedback. The system will be used to improve the performance of candidates, increase the level of accessibility, and radically change the overall environment of interview preparation.

2.3 Problem Elaboration

Our study has revealed that the current approaches and platforms of preparing interviews have a lot of challenges. The key challenges associated with these are the absence of real-time feedback, the failure to consider non-verbal communication, and language barriers. The following are some of the problems that are going to be addressed by our system in the current scenario of interview preparation:

2.3.1 Lack of Real-Time Feedback

The majority of the interview preparation techniques nowadays only give feedback following the session. This time lag does not allow candidates to immediately fix errors and the effectiveness of the

practice is lower. Nexora overcomes this drawback by providing real-time visual prompts in the session and, therefore, enabling candidates to change and get better on the fly. The behavioral feedback is instantaneous; however, the system does not send the transcript to the LLM model to evaluate the content until an answer is finished which is detected either through short silence, or when the candidate clicks Next. This analysis is then recorded in the database so that it can be used later to generate performance report.

2.3.2 Ignoring Non-Verbal Communication

The current platforms mainly concentrate on verbal or text response and seldom analyze non-verbal communication like body language, facial expression or eye engagement. Such cues are very important during interviews. To include such non-verbal cues in the performance appraisal, Nexora uses AI-powered audio and video analysis.

2.3.3 Language Barriers

The majority of platforms are monolingual and tend to have only English. This poses a problem of accessibility to candidates of different backgrounds. Nexora presents the concept of multilingual support (English, Urdu, French, Spanish) which means inclusiveness and broader adoption.

2.3.4 Limited Engagement

Conventional approaches have been traditional and non-interactive hence lowering the level of engagement among candidates. Nexora makes the preparation process more interactive, making it an AI interviewer, and the questions are asked dynamically and offer an engaging and interactive experience.

2.4 Goals and Objectives

The main idea of the platform is to redefine the process of preparing the interviews making it more interactive, transparent, and effective. It aims at giving the applicants real time instructions on both verbal and non verbal communication hence enhancing their confidence once they are faced with actual interviews.

- Provide real-time feedback to candidates during simulated interview situations.
- Design and implement a user friendly website that can afford easy and smooth interview preparation.
- Use LLM and computer vision models to examine both verbal and non-verbal communication to

have a holistic analysis.

- Incorporate multilingualism (English, Urdu, French, Spanish) to increase inclusivity.
- Change the interview preparedness by closing the gap between the traditional approaches and AI-based solutions.

2.5 Project Scope

Our team will be developing a full-fledged, end-to-end website, which will allow the candidates to take part in an AI-based simulated interview and provide them with real-time feedback on their performance. The system will act as an AI interviewer and ask questions, analyse answers and guide the candidates using visual aids. The site is also programmed to accommodate different languages making it accessible to a wide variety of users.

The system will enable users to:

- Sign up and enroll in the platform.
- Take part in simulated interviews based on AI.
- Get immediate feedback (green-proper behaviour, red-distracting behaviour).
- Improve verbal and non-verbal communication.
- Choose their own language (English, Urdu, French, Spanish).
- Have an interactive and easy to use interface in the interview preparation.

The audio and video analysis modules will be implemented using AI, NLP and computer vision techniques, primarily in Python, and will be deployed on a separate server. The results predicted will be accessed through REST APIs to the main platform. The technical architecture consists of React.js as the frontend, Ruby on Rails (RoR) as the backend, and PostgreSQL as the database. Project management will follow Agile principles using the SCRUM methodology.

The final project deliverables will be:

- An effective web-based solution that supports AI-supported interview preparation.
- User manual on how to use the application.
- Records on design, implementation, testing and evaluation process.

The project scope is specifically excluded to:

- Hosting or enterprise-level server infrastructure support of the application.

- Mobile application (iOS or Android) development.
- 3rd party HR / Recruitment integration.

2.6 Sustainable Development Goal (SDG)

Our initiative focuses on the sustainable development goal that is SDG 8, which is titled as Decent Work and Economic Growth. Our vision is to transform the process of interview preparation by implementing new, non-discriminative technologies. The system aims to enhance candidates' employability, self-efficacy and likelihood of success in interviews by analyzing all types of communicative signals in real time. The Nexora platform is designed as a preparation support tool to help develop a more fair and efficient interview readiness process, ultimately leading to supporting professional and economic development and building a skilled workforce.

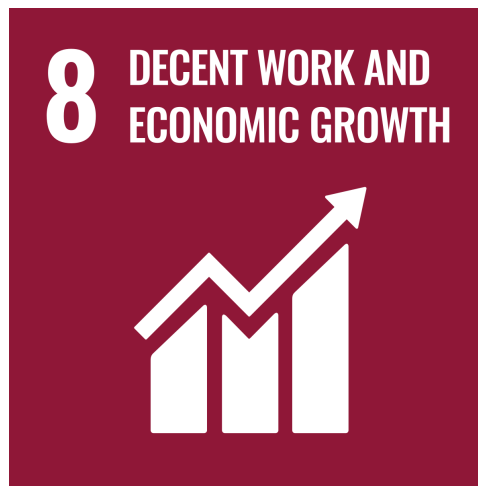


Figure 2.1: Sustainable Development Goal

2.7 Constraints

For a system like Nexora to be introduced, there are several constraints that need to be considered:

- The accuracy and responsiveness of the application heavily rely on the performance of the underlying AI and computer vision models. Any limitation in detecting facial expressions, gestures, or tone could affect the quality of real-time feedback.
- The availability of high-quality hardware such as webcams, microphones, and stable internet connections is essential. Users with poor device performance or unstable connections may experience delays or reduced accuracy in analysis.
- The AI-driven interview process may face skepticism or hesitation from traditional candidates and

institutions accustomed to conventional interview preparation methods.

- The system's processing load could increase significantly with multiple users performing real-time interviews simultaneously, which may require cloud-based scaling and optimization.
- Continuous AI model updates and retraining are needed to maintain accuracy and inclusivity across different languages, accents, and cultural variations.

2.8 Business Opportunity

The suggested product has the prospects of becoming a unique business venture. The revenue model of the company may be designed in the following way: the candidates will be free to use the basic interview practice, with fees charged to the more advanced features, including full-fledged AI-based analysis, extended practice, and multilingual support. At the same time, Nexora can be oriented to academic organizations, vocational training organizations and human-resources consultancies through the subscription model, and thus provide learners and employees with preparation tools specific to real-life interviewing.

Moreover, the solution can be bundled and sold as a recruitment-agency and corporate human-resource division software-tool to automate the process of screening candidates. Nexora has real benefits in big organisations that carry out high levels of interview operations: the savings of time, standard procedure of evaluations, and increased impartiality of assessments. As a result, the implementation of this platform will probably increase the efficiency of the recruitment processes in general and provide the candidates with a more transparent and engaging interviewing process.

2.9 Stakeholders Description/ User Characteristics

The two main stakeholders of the system are the Candidates i.e. those who prepare themselves to attend the interview and the AI System which is the interviewer.

2.9.1 Stakeholders Summary

The major users of the platform are candidates. They are provided with an individual dashboard and they can register and choose a language of their choice and start an interview session with AI. During the session, the candidates will communicate with the AI interviewer, answer the questions asked, and get visual feedback instantly aimed at improving their performance. The content of every response is assessed upon completion of answer and is stored in the database along with transcript and feedback and then used to create an integrated report on performance.

The AI system acts as the interviewer, asking questions, processing audio and video data, interpreting verbal and non-verbal messages, and providing real-time feedback to direct candidates. This creates a seamless automated process that relies on no human administrator.

2.9.2 Key High-Level Goals and Problems of Stakeholders

Nexora's primary objective is to overcome the drawbacks of conventional interview practice by offering an AI-based platform with instantaneous feedback, automation, and support in various languages. As a result, the procedure becomes more efficient, equitable, and improves the overall communication ability, providing a more convenient and individualized process.

2.9.2.1 Efficiency

This mechanism allows candidates to experience seamless interview preparation; it delivers immediate feedback, and this eliminates the delays in assessment succeeding an interview. Moreover, the candidates can do immediate behavioral changes if their positive or negative marking is immediately notified to them.

2.9.2.2 Automating the Process

The system automatically organizes the entire process of interview preparation. The candidates require just to log in and begin a session, and the AI interviewer deals with all the interview's questioning, analyzing replies, and providing real-time feedback. Thus, this obviates the necessity of human evaluators during practice sessions.

2.9.2.3 Improve Communication Skills

Being a salient component of social evolution, communication must be effective. Among the qualities of people that help them improve in social development is that of relating well with other people. Nexora may assist applicants in enhancing their verbal and non-verbal communication ability. The system offers the types of instructional support that enable an improved performance through monitoring the speech patterns, tonal quality, facial expressions, and gestures with the availability of precise guidance to the person.

2.9.2.4 Inclusivity Through Multilingual Support

The platform is also comprehensive since this is multilingual, meaning it has the capability to support such languages as English, Urdu, French, and Spanish. This will give us access to users from varying cultural backgrounds who will be able to undergo an AI-based interview preparation course.

2.10 Conclusion

To conclude, the aim of this project is to enable a better, more efficient and user-friendly interview preparation through an interactive approach. We are planning to address the challenge of feedback of interviews conducted by AI by giving real-time evaluation. The goal of this project is to help the course of real-life interviews run more smoothly by teaching candidates proper communication skills and building their confidence from preparing through simulated interviews. The project scope of Nexora requires building a complete web-based application system that merges audio and video analysis, gives immediate feedback, and supports multiple languages, which will make the solution capable and effective for all users. Non-verbal messages are provided as they happen; on the other hand, the content of the respondent would be rated after he/she has responded, and stored in the database system, thus making the system generate a relevant and precise final report on the individual candidate.

Chapter 3

Literature Review / Related Work

This chapter explores the existing research, systems, and technological approaches that influenced the creation of Nexora. A thorough investigation of earlier work in the areas of real-time interview feedback, multimodal behavioral analytics, mock interview platforms, and the integration of artificial intelligence into candidate training served as the foundation for Nexora. We aim to expand upon and improve the ideas and insights uncovered in the following works to deliver a more holistic, inclusive, and interactive interview-readiness experience.

3.1 Definitions, Acronyms, and Abbreviations

To ensure a clear understanding of the terminology used throughout the chapter, a list of acronyms and their respective meanings are provided in the list below:

AI : Artificial Intelligence

ATS : Applicant Tracking System

NLP : Natural Language Processing

HR : Human Resources

UI : User Interface

UX : User Experience

IT : Information Technology

Gen-AI : Generative Artificial Intelligence

3.2 Detailed Literature Review

The following sections provide each review material in detail including its summary, the critical analysis and relation to our work.

3.2.1 HireVue

HireVue is a proven commercial AI-based video interviewing service that has transformed the contemporary hiring practices [1]. It was established in 2004 and provides organisations with the means to interview and record and analyze interviews of candidates with the help of AI and video analytics. HireVue uses machine learning algorithms, computer vision, and NLP to evaluate both verbal and non-verbal communication, including tone, expressions, and so forth, and enables recruiters to make data-driven hiring decisions and lets the candidates have the freedom of completing interviews at their convenience remotely.

3.2.1.1 Critical Analysis

Strengths: Strong enterprise integration (ATS, workflows), polished UI/UX, end-to-end hiring features (video + assessments), and the scale/validation resources of a commercial vendor.

Weaknesses: Documented ethical and legal scrutiny around bias in automated assessments and opaque AI scoring, which has prompted audits and regulatory attention; also tends to prioritize employer needs over candidate coaching. Transparency and fairness remain recurring criticisms.

3.2.1.2 Relationship to the Proposed Work

Whereas HireVue is concentrated on the recruiters to effectively evaluate the candidates, Nexora is concentrated on improving the experience of the candidate by providing real-time feedback. On the one hand, Nexora will provide visual and auditory feedback to candidates during the interview, in contrast to HireVue, which is focused on post-interview analytics. This makes Nexora a learning and self-improvement tool and not a hiring-decision platform, which makes it a complementary innovation and not a direct competitor in the recruitment market.

3.2.2 Huru.ai

Huru.ai [2] is a real-time, AI-powered interview preparation platform that helps candidates enhance both their verbal and non-verbal communication skills. It uses facial recognition, the tone of the voice, and natural language processing to analyze the performance of the users in mock interviews. It offers

instant feedback on different aspects, like self-confidence, clarity of speech, speed of talking, and facial expressions. Huru is available via a website and a mobile app, providing an interactive platform where users improve their interview skills continuously with personal coaching.

3.2.2.1 Critical Analysis

Strengths: It is multimodal analysis with audio/video inputs for complete interview feedback. It is also real-time feedback that allows for behavioral modification of the candidate. It is also available on the web and mobile, therefore accessible to candidates of different backgrounds.

Weaknesses: It has non-verbal feedback confined to the surface level of emotion and facial recognition, without deep analysis of gestures/postures and eye contact. Its feedback is not always contextual oriented and generic at times, it does not offer multi-lingual platforms to facilitate those who are non-English applicants.

3.2.2.2 Relationship to the Proposed Work

Like Huru, Nexora was also set out to improve candidate interviews in real-time with multimodal feedback. However, Nexora adds further capabilities of introducing visual cues that use instant feedback, with green or red indicators showing the appropriateness of candidate behavior. Nexora also adds a multilingual platform and in-depth behavioral analysis of gestures and eye engagement for inclusivity. Huru.ai focuses on emotion and speech multimodal analysis; whereas Nexora develops this by advancing the concept of the all-in-one solution for Candidate Interview readiness for all persons of diversified cultural backgrounds.

3.2.3 Interviewing.io

Interviewing.io [3] is a pioneering mock interview platform that pairs candidates with provides a platform for qualified professionals to practice technical interviews anonymously. Founded in 2015, the site enables users to engage themselves in technical interviewing practice sessions with Engineers from tech giants such as Google, Amazon, Meta, or even an AI-based interviewer. The platform focuses primarily on technical and coding interviews, providing users with performance feedback and a data-driven breakdown of their problem-solving skills. The anonymity feature allows candidates to focus on skill improvement without the pressure of personal identification.

3.2.3.1 Critical Analysis

Strengths: The practice is highly realistic and domain specific, thus it seems to be the real interview situation. You have the opportunity to talk with senior interviewers and the entire anonymous arrangement reduces social anxiety which is a major plus. The feedback you get is tangible and related to your performance in coding and thus you can really see what you need to work on.

Weaknesses: It is largely coding based, so you do not receive much assistance with behavioral interview skills. Many of its high-touch sessions are paid on the platform making accessibility a little challenging to scale. And although the AI solves a plethora of coding issues, it does not capture nonverbal communication, so you are missing out on the entire body-language feedback.

3.2.3.2 Relationship to the Proposed Work

Interviewing.io shows the value of realistic practice and mentorship, whereas Nexora looks at it in a different light, i.e., providing immediate multimodal, nonverbal and verbal feedback on gestures, posture, filler words, etc. Nexora is a product based on the idea of simulated interviews, but it adds automated real-time feedback, which is generated through AI-based visual and auditory analysis. Unlike interviewing.io that requires human mentors, Nexora provides real-time, continuous insights without the need to have external assessors. In addition, Nexora goes further by examining gestures, facial expression, and vocal tone, areas that interviewing.io does not cover, which makes it a more in-depth tool of interview preparation in a wide range of interviewing scenarios.

3.2.4 Google-Interview warmup

The interview preparation tool Google-Interview warmup [4] is an AI-driven interview preparation tool created by Google, and available via the Grow with Google initiative. The tool enables the practice of common interview questions in various professional fields including data analytics, UX design, and IT support. The platform uses speech recognition and natural language processing methods to transcribe the verbal answers of applicants and provide them with some analytical feedback on lexical choices, points of salient discussion, and prosodic patterns. It is supposed to allow the interviewees to critique their answers and to improve their communicative self-confidence based on the expectations of real interview scenarios. It is focused on general interviewing preparation and offers a judgment-free practice environment without the heavy tech stack.

3.2.4.1 Critical Analysis

Strengths: It is free, you can hit it anywhere and the live transcript and basic feedback makes it very easy to use. Supported by the brand of Google, and integrated into career certs, it is valid.

Weaknesses: The feedback is primarily written; no non-verbal interpretation of gestures, posture or eye contact. You are not provided with the real-time visual answers that a more advanced system would provide.

3.2.4.2 Relationship to the Proposed Work

Whereas Google Interview Warmup demonstrates the usefulness of live transcription and simple practice, Nexora takes it a step further by providing a multimodal, real-time, fully interactive coaching experience. Not only does Nexora support live transcription and NLP evaluation (fluency, filler words, etc.), but also provides computer-vision analysis of gestures, facial expression, and eye contact, which Google tool is incapable of. It also offers visual cues (green/red indicators) in real-time to prompt the candidates to act immediately, as well as multilingual approach to appeal to a larger audience. This is why Nexora is a next-generation, holistic interview preparation tool, which is based on the principles of other applications such as Interview Warmup, yet provides more detailed, practical, and inclusive feedback.

3.2.5 Rezi.ai

Rezi.ai [5] is an artificial intelligence-based mock interviewing website that is also linked with the Rezi resume platform. It tailors interview questions based on the candidate's resume and the specific job description they are targeting. The system captures responses, transcribes them and provides elaborate post response reviews in terms of conciseness, use of keywords and clarity of the answers. Rezi.ai makes the preparation very contextual and job-oriented by matching the interview. Practice with a background of a candidate and the role he is targeting. .

3.2.5.1 Critical Analysis

Strengths: It easily integrates with users' resumes and job postings for customized tailored interview questions specific to a candidate's context. It provides feedback after every answer. On conciseness, keyword usage, and overall impression-all helpful to the candidate in fine-tuning their responses for better alignment with job expectations. Accessible online, easy to set up for wide application.

Weaknesses: Responses are provided as feedback after the answers and do not provide any real-time, on-the-interview simulation guidance. Only audio/text analysis, without monitoring non-verbal behaviors pertaining to gesture, posture, or eye contact. Lacks multilingual support and wider adaptability for a diverse cultural or linguistic set of users. The feedback is descriptive but not interactive; hence, less effective as a continuous coaching tool.

3.2.5.2 Relationship to the Proposed Work

Nexora offers real-time, multimodal feedback, as opposed to post-response commentary in Rezi.ai. While Rezi.ai customizes questions based on the resume and the job description and evaluate answers based on key-words presence, Nexora provides on-the-fly coaching through visual signals and uses a semantic LLM to evaluate the full meaning and quality of a candidate's answer, allowing job seekers to change their behaviour in real time during the interview. Therefore, Nexora is superior to the simulation method used by Rezi.ai, which is more interactive and inclusive in terms of the training environment.

3.2.6 Yoodli.ai

Yoodli.ai [6] is a conversational AI platform that provides candidates with real-time coaching during interview practice. It examines spoken responses to determine the presence of filler words, pacing, confidence, and clarity as well as the non-verbal cues like body language. Yoodli.ai also uses role-play scenarios and dynamic follow-up questions, unlike traditional practice tools that only simulate interview conditions. It has a web-based interface and at the same time can be exercised both in private practice mode and live. Coaching mode, thus providing a wide range of preparation options.

3.2.6.1 Critical Analysis

Strengths: Provides candidates with instant feedback on pacing, speech clarity, and filler words so they can adjust their responses. provides a deeper understanding than text- or audio-only platforms by evaluating both verbal and some non-verbal behaviors. Its simulated interviews can be made more realistic and interactive by using role-play and dynamic questioning. The flexible practice modes of this web-based tool allow for a variety of candidate preparation needs.

Weaknesses: Although it takes nonverbal cues into consideration, the visual feedback is typically limited and does not take precise gestures, posture, or eye contact into account. Candidates from all over the world are less inclusive because multilingual options receive little to no support. Some feedback may remain generalized and lacks transparency in explaining why certain weaknesses are flagged. Its reliance on AI generated analysis may sometimes oversimplify nuanced aspects of candidate performance.

3.2.6.2 Relationship to the Proposed Work

Nexora has the same goal of presenting real-time interactive interview coaching, like Yoodli.ai, but extends this feature by offering a structured visual cueing system. Where Yoodli.ai focuses on speech measures and body-language analysis, Nexora is a multimodal monitoring system, covering gestures and a multilingual interface. Because of this, Nexora is a more holistic and provide inclusive coaching experience.

3.2.7 Leveraging Multimodal Behavioral Analytics for Automated Job Interview Performance Assessment and Feedback

This research paper Leveraging Multimodal Behavioral Analytics for Automated Job Interview Performance Assessment and Feedback is written by A. Agrawal et al. [7]. The authors devise a system, which can automatically grade job interviews using data gathered through three sources: audio, transcripts, and certain basic visual cues. It verifies such aspects as the speed of speech, accent, the tone of the responses, and even mere facial expressions. Then machine-learning gives out a score of performance. They demonstrate how having all three channels together will provide an improved view than one.

3.2.7.1 Critical Analysis

Strengths: This system is a perfect synchronization of audio, text, and video to provide a detailed analysis of the interview performance. It feeds data to a predictive model through feature extraction of each modality to estimate human interviewer scoring. This is a multimodal system that is a major step forward in the old systems of text or voice alone systems as it provides a more comprehensive measure of candidate performance.

Weaknesses: Although the system has an advanced analysis pipeline, the functionality of the system does not support real-time feedback, as it is only able to produce a post-interview score. The visual analysis functions are limited to it, omitting such vital elements as hand gestures, body language, and eye contact. Moreover, the system is not multilingual and fails to provide the logic behind the given scores, which makes it less transparent and less applicable to various, real-life interview situations.

3.2.7.2 Relationship to the Proposed Work

Although that paper demonstrates that multimodal analytics are workable in the interview process, Nexora takes it a step further and provides real-time, on-screen feedback. Nexora displays green/red icons

instead of post-session scores, meaning that you can see at a glance how you are doing, and it can handle full video footage, including gestures, posture, facial expressions, as well as audio and text. It also supports a variety of languages (English, Urdu, French, Spanish) which opens it down to a ton more individuals. So Nexora transforms a research prototype into an entire coaching tool that can assist you in getting better as you actually interview.

3.2.8 AI Powered Mock Interview System with Real-Time Voice and Emotion Analysis

This research paper AI Powered Mock Interview System with Real-Time Voice and Emotion Analysis is written by S. Khapekar et al. [8]. The article presents a simulated interview system, which does real-time voice feature and emotional tone analysis. It tracks the pitch, tone, and emotional condition of candidates during mock interviews and offers an audio-based feedback instantly to allow them to improve their vocal delivery, decrease nervousness, and manage emotions under stress.

3.2.8.1 Critical Analysis

Strengths: The research article contributes to the study of interactive interview preparation because it shows how the use of real-time voice and emotion feedback can improve the performance of job applicants during the practice. Multitasking of instant auditory and emotional analysis allows the candidate to find the improvement areas in a short time, which fosters self-awareness and confidence in communication.

Weaknesses: Although the platform is innovative, it is one-dimensional as it requires only audio and emotional contributions, and it overlooks such a crucial non-verbal visual element as gestures, posture, and eye contact. Moreover, it does not have a multilingual interface, which limits its availability to different users. The feedback given is also divided, it points out the single issues but does not produce a multimodal coaching experience that incorporates all the elements of interview performance.

3.2.8.2 Relationship to the Proposed Work

Research article confirm the effectiveness of real-time audio/emotion feedback, but Nexora goes a step further and develops a complete multimodal coaching experience. Nexora not only analyzes voice and video (gestures, posture, facial expression) simultaneously, but also uses instant visual cues (green/red indicators), and is multi-lingual. This makes a one-channel mock interview tool an inclusive and transparent real time coaching system which directly increases candidate performance in the interview.

3.2.9 Virtual Self-Practice Mock Interview Using Gen-AI

This research paper Virtual Self-Practice Mock Interview Using Gen-AI is written by M. Suguna et al. [9]. The authors propose a self-practice mock interview platform driven by generative AI. It basically allows you to rehearse interview questions and answers without the need to have a real interviewer. The system provides audio or written feedback to you regarding your responses, the aim being to enable you to improve your responses and to reflect on them through much practice.

3.2.9.1 Critical Analysis

Strengths: The article is very helpful in pointing out the possibility of using generative AI to generate personalized and dynamic interview situations. This will make it scalable and accessible so that more people can work with the interviews that are as close to reality as possible. The flexibility of generative AI also means that the system can adapt questions and the level of difficulty to the individual performance, which will give the interview simulation experience a personal approach.

Weaknesses: Although these benefits are present, the feedback provided by the system is mostly audio or text-based and is not delivered until the responses are provided. It is not as interactive and responsive because it is not analyzed in real-time, non-verbally, and does not provide instant visual feedback. Also, it lacks multilingual support making it less inclusive to users who have a different language. The experience is therefore more of a simulated set-up instead of an interactive coaching session that encourages live learning and behavior change.

3.2.9.2 Relationship to the Proposed Work

The paper demonstrates the possibilities of Gen-AI regarding scalable mock interviews. Nexora goes a step further and combines generative NLP with multimodal and real-time coaching. It also introduces computer-vision-based analysis, visual red/green cues, and multilingual support making a passive self-practice tool an active, interactive and inclusive interview-readiness tool.

3.3 Literature Review Summary Table

The related applications and research summary are shown in Table 3.1 and Table 3.2.

Table 3.1: Summary of related applications

Application	Features	Relevance	Limitations
HireVue [1]	AI-powered video interviewing platform using NLP, facial expression analysis, and automated scoring	Demonstrates the use of AI for analyzing verbal and non-verbal behavior in interviews	Primarily benefits employers; limited real-time feedback for candidates; potential algorithmic bias
Huru.ai [2]	AI interview preparation platform analyzing facial emotions, tone, and speech using NLP and computer vision	Shows the potential of multimodal, real-time feedback systems for candidate self-improvement	Limited to basic emotion and speech analysis; lacks full gesture/posture tracking and multilingual capability; feedback can be generic
Interviewing.io [3]	Mock interview platform connecting users with real engineers for live technical interviews	Highlights the importance of realistic simulations and performance-based learning	Limited to technical interviews; relies on human feedback; excludes non-verbal analysis
Google Interview Warmup [4]	Free AI tool for practicing interview questions using speech recognition and NLP	Shows effectiveness of AI in linguistic feedback and self-paced interview preparation	Analyzes only textual and verbal responses; lacks gesture, tone, or posture analysis
Rezi.ai [5]	Tailors interview questions based on resume/job description; provides post-response evaluation of clarity, conciseness, and keyword usage	Demonstrates the value of customized, job-specific mock interviews aligned with candidate applications	Feedback only post-response; lacks real-time guidance, multimodal analysis, and multilingual support
Yoodli.ai [6]	Provides real-time coaching with nudges on pacing, filler words, and deliveries	Highlights the potential of interactive, real-time feedback and dynamic questioning in interview prep	Visual analysis limited (gestures/eye contact not fully captured); multilingual support unclear; feedback may lack transparency

Table 3.2: Summary of related research

Authors	Features	Relevance	Limitations
Agrawal et al. (2020) [7]	Multimodal behavioral analytics framework combining audio, transcripts, limited visual cues for post-hoc interview scoring	Proves feasibility of combining modalities to assess interview performance, inspiring Nexora's approach	Post-hoc evaluation only, limited visual analysis, and no real-time candidate feedback
Khapekar et al. (2025) [8]	Real-time voice and emotion analysis in mock interviews	Validates benefit of real-time audio/emotion feedback which Nexora expands upon	Audio/emotion only, lacks full video gesture/posture analysis and multilingual support
Suguna et al. (2024) [9]	Generative AI for self-practice mock interviews, simulating interviewer responses and providing audio/textual feedback	Shows potential of Gen-AI for scalable mock interviews that Nexora can extend	Feedback mainly post-response, lacks multimodal analysis, no instant visual cueing or multilingual interface

3.4 Conclusion

In conclusion, this chapter has discussed the complex interrelations between Nexora and some of the most popular interview-preparation systems and research projects, i.e., HireVue, Huru.ai, Interviewing.io, Google Interview Warmup, Rezi.ai, Yoodli.ai, and scholarly works by Agrawal et al., Khapekar et al., and Suguna et al. All these platforms and papers have their peculiarities and ideas that can be used to inform and inspire the development of Nexora. Nexora will focus on the strengths and weaknesses of these methods to use the most effective ones, namely real-time analytics, multimodal feedback, and scalable practice, and fill the gaps that they leave. In this, Nexora aims to provide a holistic, inclusive, and interactive interview-readiness platform that establishes a new standard in coaching and performance enhancement of candidates.

Chapter 4

Software Requirement Specifications

This chapter provides a comprehensive analysis of all software requirements associated with the Nexora system. It includes a detailed list of features, functional and non-functional requirements, quality attributes, and assumptions made during system design. Furthermore, it outlines the hardware and software requirements, use cases, graphical user interfaces, and all relevant diagrams that illustrate the system's structure and behavior. Finally, the chapter concludes with the data dictionary and risk analysis, providing a complete understanding of the functional foundation of Nexora.

4.1 List of Features

It's not just an app for conducting mock interviews; rather, it's an AI-powered coach that turns every practice session into a sprint of improvement. The following are some of the features of Nexora that make it a job-seeker's game-changer:

1. **Secure User Authentication:** Secure login and registration are provided, allowing users to safely access their accounts.
2. **AI-Driven Mock Interviews:** Carries out AI-supported mock interviews that mirror authentic job employment situations in order to assist candidates in hypothetical employment. The system poses questions that are applicable to the discipline, pushes the interview process along naturally, and provides feedback as the interview occurs to facilitate communication and performance.
3. **Real-Time Feedback:** Provides immediate visual feedback through indicators such as colored bars or icons that reflect tone, eye contact, and gestures. This helps the candidate maintain proper engagement and adjust their interview behavior in real time.
4. **Performance Analytics:** Provides a comprehensive performance report after every session about

strengths, areas of improvement, and overall scores.

5. **AI Improvement Suggestions:** Provides personalized coaching suggestions using machine learning models to enhance user performance.
6. **Profile Management:** Allows users to update personal details, contact information, and profile picture.
7. **Language Customization:** Enables users to select their preferred language for AI interaction to support diverse users.
8. **Pause and Resume Interviews:** Allows users to pause an ongoing interview and resume later from the same point without losing progress.
9. **LLM-Based Answer Evaluation:** The system evaluates the content of every answer with a semantic LLM as soon the user finish its response. The assessment is recorded to a database and used for generating the overall report, preserving the continuity of interview process.

4.2 Functional Requirements

This section explains what the Nexora system should be able to do. These are the main actions or features that make the system useful and complete.

- The system shall allow the user to start a mock interview after successful authentication.
- The system shall enable real-time interaction between the user and the AI interviewer.
- The system shall provide instant feedback during the interview using color indicators or short comments.
- The system shall evaluate the content of each answer only after the user finishes speaking or presses Next, using a semantic LLM.
- The system shall store each evaluated answer in the database for later reporting.
- The system shall generate and display a detailed performance report at the end of the interview, including ratings and recommendations.
- The system shall analyze past interview data and provide personalized tips and recommendations.
- The system shall allow users to view and update their personal profile (name, email, profile photo).
- The system shall allow the user to select the language for the interview.
- The system shall allow the user to pause an ongoing interview and resume it later without losing

progress.

- The system shall securely store user data and prevent unauthorized access.
- The system shall display clear and informative error messages in cases such as invalid input or connectivity issues.

4.3 Quality Attributes

This section provides the constraints and quality attributes of the Nexora system. Requirements are expressed in terms of the performance, usability, reliability, security, and operational profile of the system rather than in functional terms.

4.3.1 Usability

The system should have a very easy and intuitive user interface that can easily be learned. The users shall be able to accomplish their required task efficiently and shall not get confused or require very extensive training. It shall include usability features such as clear labeling, walkthroughs, and helper texts to make it more user-friendly.

4.3.2 Reliability

The system should work reliably in normal scenarios without crashes. The system shall maintain data integrity and prevent any corruption of data or loss of data in operations related to interviews, profile creation, and report generation. The system shall have a plan for graceful recovery from failures.

4.3.3 Accuracy

The AI module will score user responses and give feedback that truly represents their performance. The scoring algorithm will be based on established metrics for fairness and consistency.

4.3.4 Privacy

The system shall ensure that user data, and especially personal data, are kept private and not disclosed to unauthorized parties. In this regard, no data of the users shall be disclosed to any third parties without seeking express consent from the users.

4.3.5 Security

The system shall be designed to counteract unauthorized access by using secure authentication and access control. All sensitive data shall be encrypted at rest and in transit. The system shall prevent any security breach and malicious attack.

4.3.6 User Satisfaction

The system should provide a user experience that would motivate and drive users to invest time and effort in practicing and improving their interviews. The system shall provide feedback relevant to the improvement of the user, a seamless experience with the AI interviewer, and rich toolsets that help users learn and gain confidence.

4.4 Non-Functional Requirements

This section defines how the Nexora system should perform rather than what it should do. These requirements focus on the system's quality, performance, and overall user experience.

- The system shall respond quickly to user actions, including loading pages, starting interviews, and generating reports, with acceptable response times even under high usage or network latency.
- The system shall protect all user information through secure authentication, authorization, and encrypted data storage to prevent unauthorized access.
- The system shall function correctly across major web browsers such as Chrome, Firefox, and Edge, and on multiple device types including desktop, tablet, and mobile.
- The system shall be modular and well-documented to allow easy maintenance, issue resolution, and future feature expansion.
- The system shall remain accessible with minimal downtime, and any scheduled maintenance shall be planned and communicated to users in advance.
- The system shall create secure backups of user data at regular intervals and provide reliable recovery options in case of system failure or data loss.

4.5 Assumptions

This section lists the main assumptions made during the development and design of the Nexora system. These assumptions help define the boundaries and expected conditions under which the system will operate.

- Users are expected to have a working internet connection for video interviews and feedback.
- Every participant is expected to be in possession of a participating device equipped with a camera and a microphone, hence they can use their laptop or mobile phone.
- Information provided by a user must be true and accurate to the best of one's knowledge.
- The AI interviewer module is assumed to have access to trained ML models in order to analyze the performance of users.
- Users are expected to grant access to the camera and microphone when doing their interviews.
- The server and database will remain available and secure during all system operations.

4.6 Use Cases

This section identifies major Nexora system use cases. The following use cases present key interaction scenarios.

Table 4.1: User Registration

Name	User Registration		
Actors	Candidate (User)		
Summary	The user will enter valid details to register and create an account on the Nexora platform.		
Pre-Conditions	User must not already be registered in the database. Internet connectivity must be available.		
Post-Conditions	A new user account is created and the user can log in using their credentials.		
Special Requirements	Unique email and valid password combination.		
Basic Flow			
Actor Action		System Response	
1	The user opens the signup page.	1	The signup form is displayed asking for name, email, and password.
2	The user enters the required details.	2	The system validates the entered information.
3	The user submits the signup form.	3	The system creates a new account and shows: “Account created successfully. Please log in.”
Alternative Flow			
1	The user enters an already registered email.	1	The system shows: “Email already exists. Please log in or reset password.”
2	The user enters a weak password.	2	The system shows: “Password must meet security requirements.”
3	The user leaves required fields empty.	3	The system shows: “Please fill in all required fields”

Table 4.2: User Login

Name		User Login	
Actors		Candidate (User)	
Summary		The user will enter valid credentials to access the Nexora dashboard.	
Pre-Conditions		User must already be registered in the database. Internet connectivity must be available.	
Post-Conditions		User session is created and access is granted to the Nexora dashboard.	
Special Requirements		Valid email and password combination.	
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	1	The login form is displayed asking for email and password.
2	The user enters valid credentials.	2	The system authenticates the user and opens the dashboard.
Alternative Flow			
1	Invalid email or password entered.	1	The system shows: “Incorrect email or password.”

Table 4.3: User Logout

Name	User Logout		
Actors	Candidate (User)		
Summary	The user clicks the logout option to securely end their active session and exit the Nexora platform.		
Pre-Conditions	The user must be logged into the system with an active session. Internet connectivity must be available.		
Post-Conditions	The user’s session is terminated and they are redirected to the login or home page.		
Special Requirements	Stable internet connection and active user session.		
Basic Flow			
Actor Action		System Response	
1	The user clicks the “Logout” button.	1	The system confirms logout request.
2	The user confirms logout.	2	The system terminates session and redirects to login page.
Alternative Flow			
1	Network issue occurs during logout.	1	The system displays an error: “Logout unsuccessful ”

Table 4.4: Forgot Password

Name		Forgot Password	
Actors		Candidate (User)	
Summary		The user can reset their password by requesting a reset link, which is sent to their registered email address.	
Pre-Conditions		The user must already have an account registered in the system with a valid email address.	
Post-Conditions		A password reset link is sent to the user’s email, allowing them to set a new password successfully.	
Special Requirements		Stable internet connection and valid registered email.	
Basic Flow			
	Actor Action		System Response
1	The user clicks on “Forgot Password” on the login page.	1	The system displays the password recovery form.
2	The user enters their registered email and submits.	2	The system validates the email and sends a password reset link.
Alternative Flow			
1	The user enters an invalid or un-registered email.	1	The system shows an error: “Email not found in records.”

Table 4.5: Start Mock Interview

Name		Start Mock Interview	
Actors		Candidate, AI Interviewer	
Summary		The user starts an AI-driven interview session where the system records responses and provides live analysis.	
Pre-Conditions		User must be logged in and allow access to camera and microphone.	
Post-Conditions		Mock interview session begins with real-time monitoring activated.	
Special Requirements		Stable internet connection and functional camera/microphone.	
Basic Flow			
Actor Action		System Response	
1	User clicks “Start Interview”.	1	The AI interviewer loads and begins asking questions.
2	User responds to questions.	2	The system captures and analyzes the responses.
Alternative Flow			
1	User denies camera/mic access.	1	The system shows: “Permission required to continue.”

Table 4.6: Real-Time Feedback

Name	Real-Time Feedback		
Actors	Candidate, AI Interviewer		
Summary	System analyzes verbal and non-verbal behavior during the interview and displays real-time cues.		
Pre-Conditions	Interview session must be active and media streams running.		
Post-Conditions	Feedback is continuously displayed through red/green signals.		
Special Requirements	Low-latency AI processing and smooth video stream.		
Basic Flow			
Actor Action		System Response	
1	User speaks and gestures during interview.	1	System processes audio-video feed.
2	User adjusts based on feedback.	2	System updates cues accordingly (red/green).
Alternative Flow			
1	Network lag or signal loss.	1	System pauses feedback and alerts user to check connection.

Table 4.7: Evaluate and Store Candidate's Answer

Name	Evaluate and Store Answer																								
Actors	Candidate, AI System																								
Summary	System detects answer completion, evaluates it, and stores results for final report.																								
Pre-Conditions	Active interview session and working STT.																								
Post-Conditions	Evaluation saved in database.																								
Special Requirements	Stable internet connection.																								
Basic Flow																									
<table><tr><th colspan="2">Actor Action</th><th colspan="2">System Response</th></tr><tr><td>1</td><td>User finishes answer or clicks “Next”.</td><td>1</td><td>System Detects end of answer.</td></tr><tr><td>2</td><td>—</td><td>2</td><td>System generates the answer transcript.</td></tr><tr><td>3</td><td>—</td><td>3</td><td>System sends the transcript to the LLM.</td></tr><tr><td>4</td><td>—</td><td>4</td><td>System receives the evaluation results.</td></tr><tr><td>5</td><td>—</td><td>5</td><td>System saves the evaluation in the database.</td></tr></table>		Actor Action		System Response		1	User finishes answer or clicks “Next”.	1	System Detects end of answer.	2	—	2	System generates the answer transcript.	3	—	3	System sends the transcript to the LLM.	4	—	4	System receives the evaluation results.	5	—	5	System saves the evaluation in the database.
Actor Action		System Response																							
1	User finishes answer or clicks “Next”.	1	System Detects end of answer.																						
2	—	2	System generates the answer transcript.																						
3	—	3	System sends the transcript to the LLM.																						
4	—	4	System receives the evaluation results.																						
5	—	5	System saves the evaluation in the database.																						
Alternative Flow																									
<table><tr><td>1</td><td>LLM timeout.</td><td>1</td><td>System retries or marks evaluation as pending.</td></tr><tr><td>2</td><td>Network issue.</td><td>2</td><td>System queues the evaluation</td></tr></table>		1	LLM timeout.	1	System retries or marks evaluation as pending.	2	Network issue.	2	System queues the evaluation																
1	LLM timeout.	1	System retries or marks evaluation as pending.																						
2	Network issue.	2	System queues the evaluation																						

Table 4.8: View Performance Report

Name		View Performance Report	
Actors		Candidate	
Summary		Candidate sees report with scores and recommendations	
Pre-Conditions		Interview completed	
Post-Conditions		Report displayed with actionable suggestions	
Special Requirements		Secure access to user-specific data	
Basic Flow			
Actor Action		System Response	
1	Interview ends.	1	System displays report with scores and recommendations.
2	Candidate views report.	2	System highlights key points and suggestions.
Alternative Flow			
1	No interview completed.	1	System displays message: “No report available.”

Table 4.9: Update Profile

Name	Update Profile		
Actors	Candidate		
Summary	User updates personal details such as name, email, or profile photo in the Nexora system.		
Pre-Conditions	User must be logged in.		
Post-Conditions	Updated details are saved and reflected in the user profile.		
Special Requirements	Proper validation and secure data storage.		
Basic Flow			
Actor Action		System Response	
1	User navigates to “Profile Settings”.	1	System displays editable profile information.
2	User edits and saves changes.	2	System validates and updates profile data.
Alternative Flow			
1	Invalid data entered.	1	System displays error: “Invalid input.”

Table 4.10: Change Interview Language

Name	Change Interview Language		
Actors	Candidate		
Summary	User selects preferred interview language before the session begins to enable localized questions and responses.		
Pre-Conditions	User must have an active account and upcoming interview.		
Post-Conditions	Interview session is set in the chosen language.		
Special Requirements	Language model and translation APIs must be available.		
Basic Flow			
Actor Action		System Response	
1	User opens “Settings”.	1	System shows available languages.
2	User selects a language.	2	System conducts interview accordingly.
Alternative Flow			
1	Selected language not supported.	1	System shows error message.

Table 4.11: Pause/Resume Interview

Name	Pause/Resume Interview		
Actors	Candidate, AI Interviewer		
Summary	User can pause and later resume the ongoing interview without losing session data.		
Pre-Conditions	An active mock interview must be in progress.		
Post-Conditions	Session resumes from the exact paused state.		
Special Requirements	Session buffering and data persistence required.		
Basic Flow			
Actor Action		System Response	
1	User clicks “Pause”.	1	System freezes interview temporarily.
2	User clicks “Resume”.	2	System restarts interview from saved point.
Alternative Flow			
1	User exits during pause.	1	System saves progress and ends session.

Table 4.12: Share Performance Report

Name	Share Performance Report		
Actors	Candidate		
Summary	User shares their performance report with mentors or peers via a secure generated link or email.		
Pre-Conditions	At least one completed performance report must be available.		
Post-Conditions	A secure shareable link or email is successfully generated and sent.		
Special Requirements	Proper email configuration and secure link encryption required.		
Basic Flow			
Actor Action		System Response	
1	User opens “Reports” section.	1	System displays available reports.
2	User selects “Share Report”.	2	System generates secure link or email option.
3	User confirms sharing.	3	System sends link or email confirmation.
Alternative Flow			
1	Network error during sending.	1	System shows message: “Failed to share report.”

4.7 Hardware and Software Requirements

This section lists the hardware and software needed to develop, test, and use the Nexora system. These requirements make sure the system works smoothly and efficiently.

4.7.1 Hardware Requirements

The following hardware components are needed for both the development team and end users.

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB (16 GB recommended for developers)
- **Storage:** At least 256 GB SSD for fast performance
- **Internet Connection:** Stable broadband connection for real-time feedback and video interviews
- **Camera and Microphone:** Required for conducting and recording mock interviews
- **Display:** Minimum 14-inch screen with 1080p resolution

4.7.2 Software Requirements

The following software tools and platforms are used to build and run the Nexora system.

- **Operating System:** Windows 10/11, macOS, or Linux
- **Backend Framework:** Ruby on Rails
- **Frontend Framework:** React.js , JavaScript
- **Database:** PostgreSQL
- **AI and Machine Learning Tools:** Python (TensorFlow / OpenCV for AI feedback and emotion analysis)
- **Version Control:** Git and GitHub for code management

4.8 Graphical User Interface

This section displays GUI screenshots of Nexora.

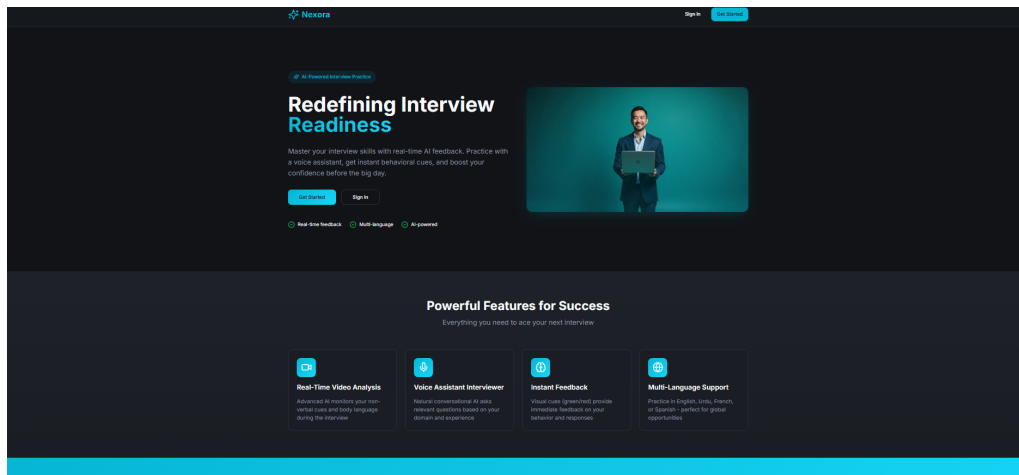


Figure 4.1: Landing Page

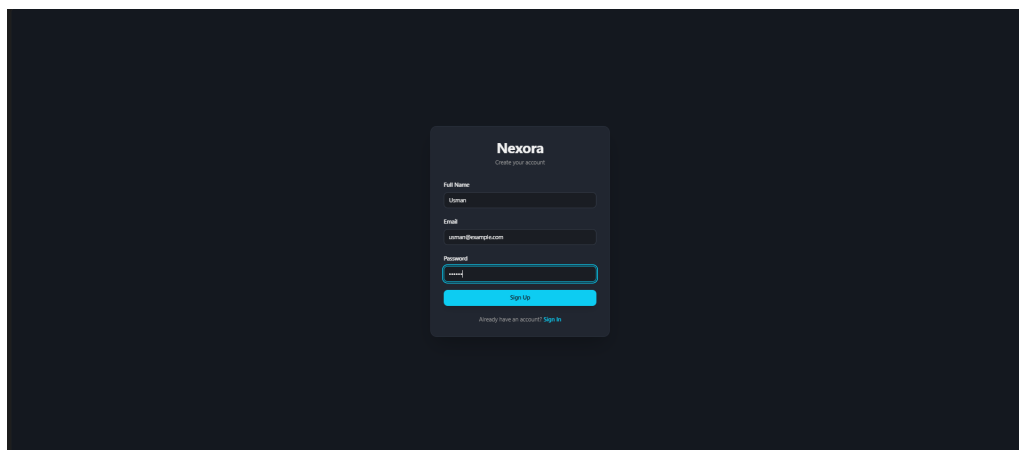


Figure 4.2: Signup Screen

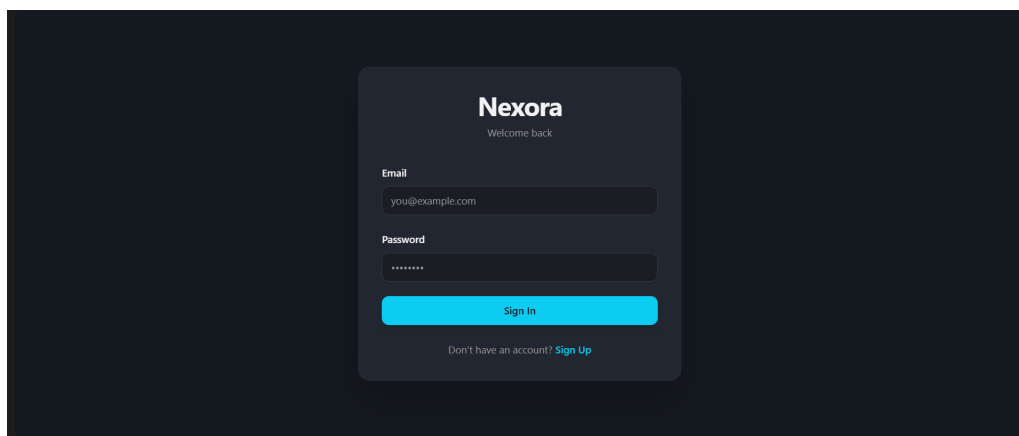


Figure 4.3: Signin Screen

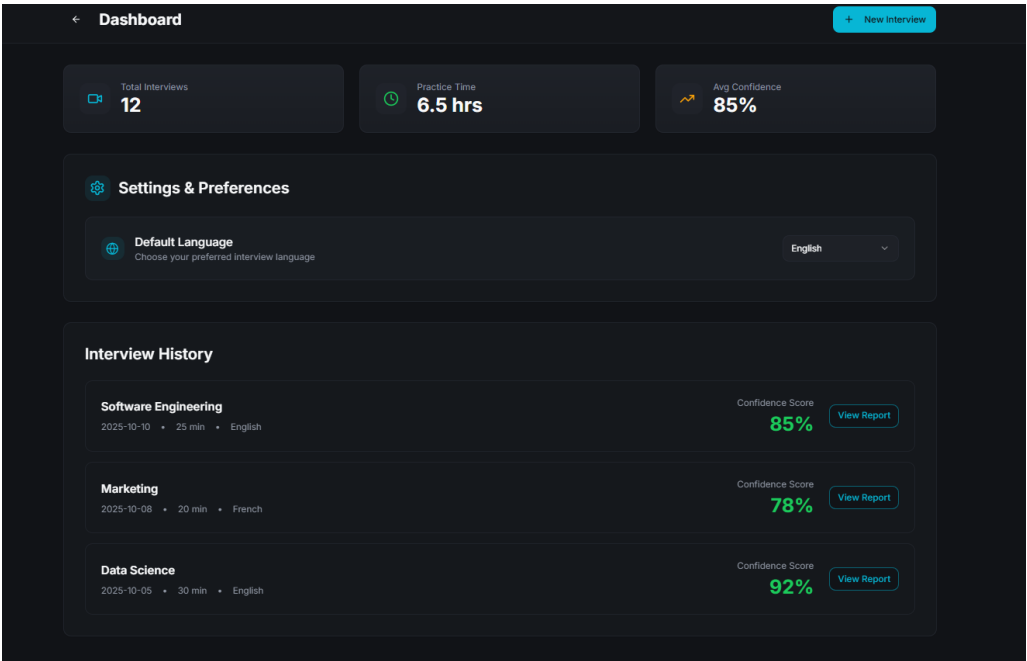


Figure 4.4: Dashboard Screen

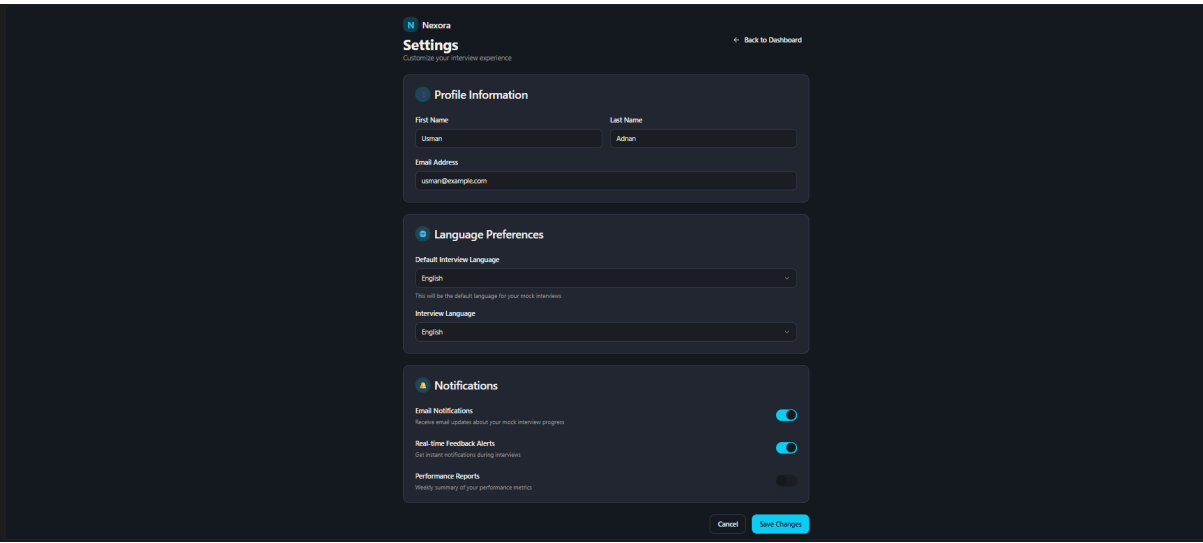


Figure 4.5: User Settings Page

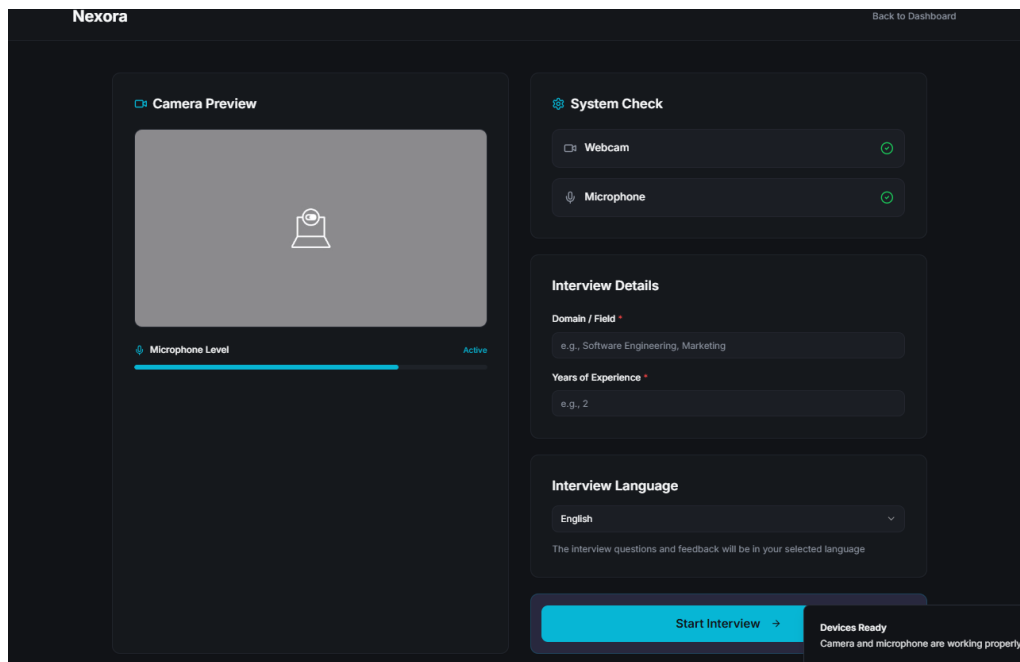


Figure 4.6: Interview Setup Screen

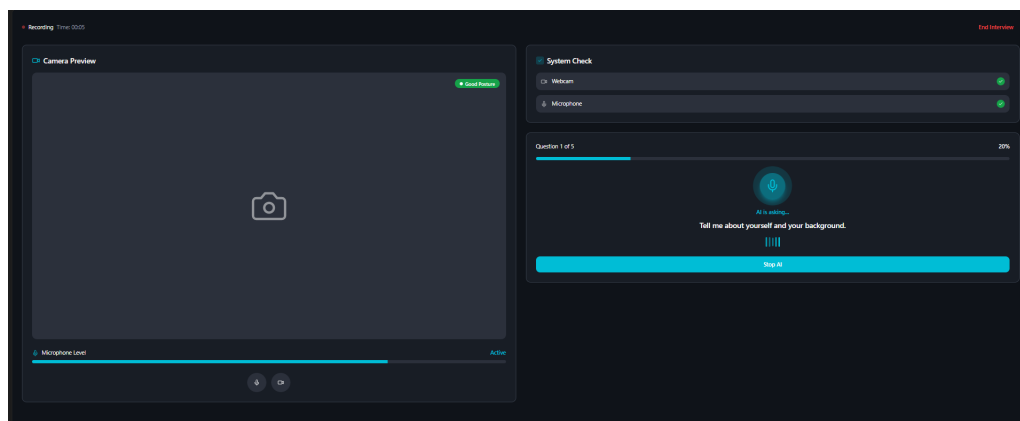


Figure 4.7: Interview Screen

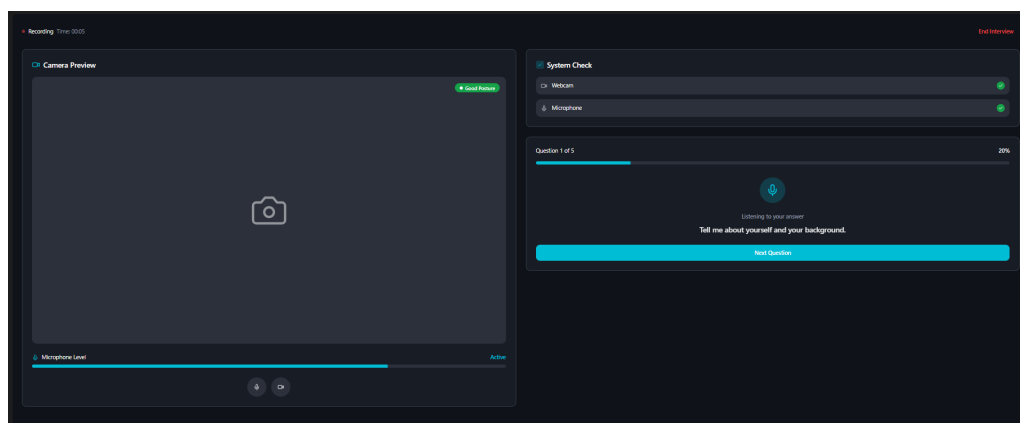


Figure 4.8: User Response Screen

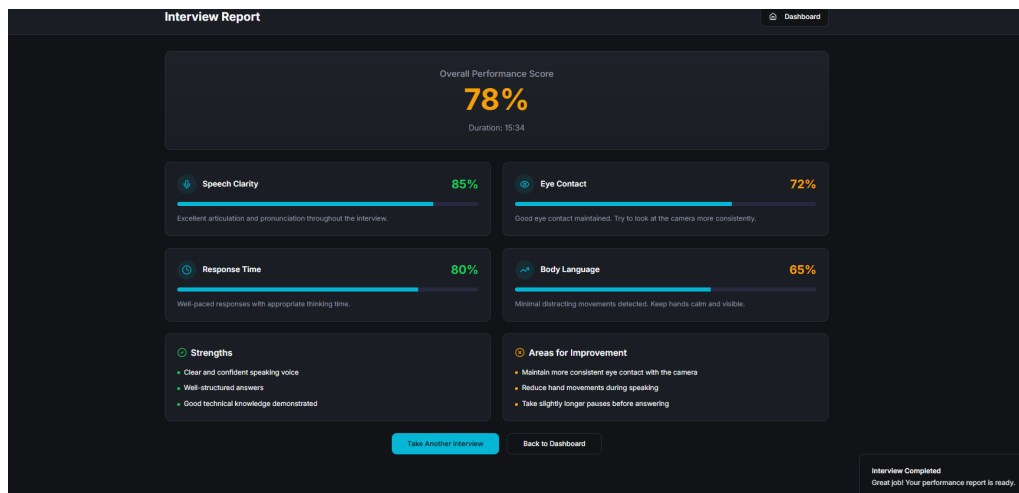


Figure 4.9: Generated Interview Performance Report Screen

4.9 Database Design

The Entity Relationship Diagram in Section 4.9 shows the interactions between all the fundamental elements of the Nexora system to support user management, interview sessions, and AI-generated feedback. The central component of the design is the User entity which contains crucial authentication and identity related details. A User Profile is the association of each user, and is in a one to one relationship where extended information like the contact number and profile photo of the user is kept. Users can attend several interview sessions which is represented by Interview entity. An interview is associated with a certain Language and the system supports the multilingual interviews, and an interview has its present status, including active or finished. In an interview, the system captures the individual Responses which are the questions and the answers given by the candidate and this forms a one to many relationship between Interview and Response. After an interview is over, the system produces a Feedback Report, uniquely associated with that interview and including the measures of verbal and non-verbal scores, overall performance, and AI-provided recommendations. Overall, it is possible to state that the ERD offers a clear and systematic overview of how user data, interview sessions, and AI evaluation results are organized in the database, which provides consistency, integrity, and effective data flow throughout the system.

4.9.1 ER Diagram

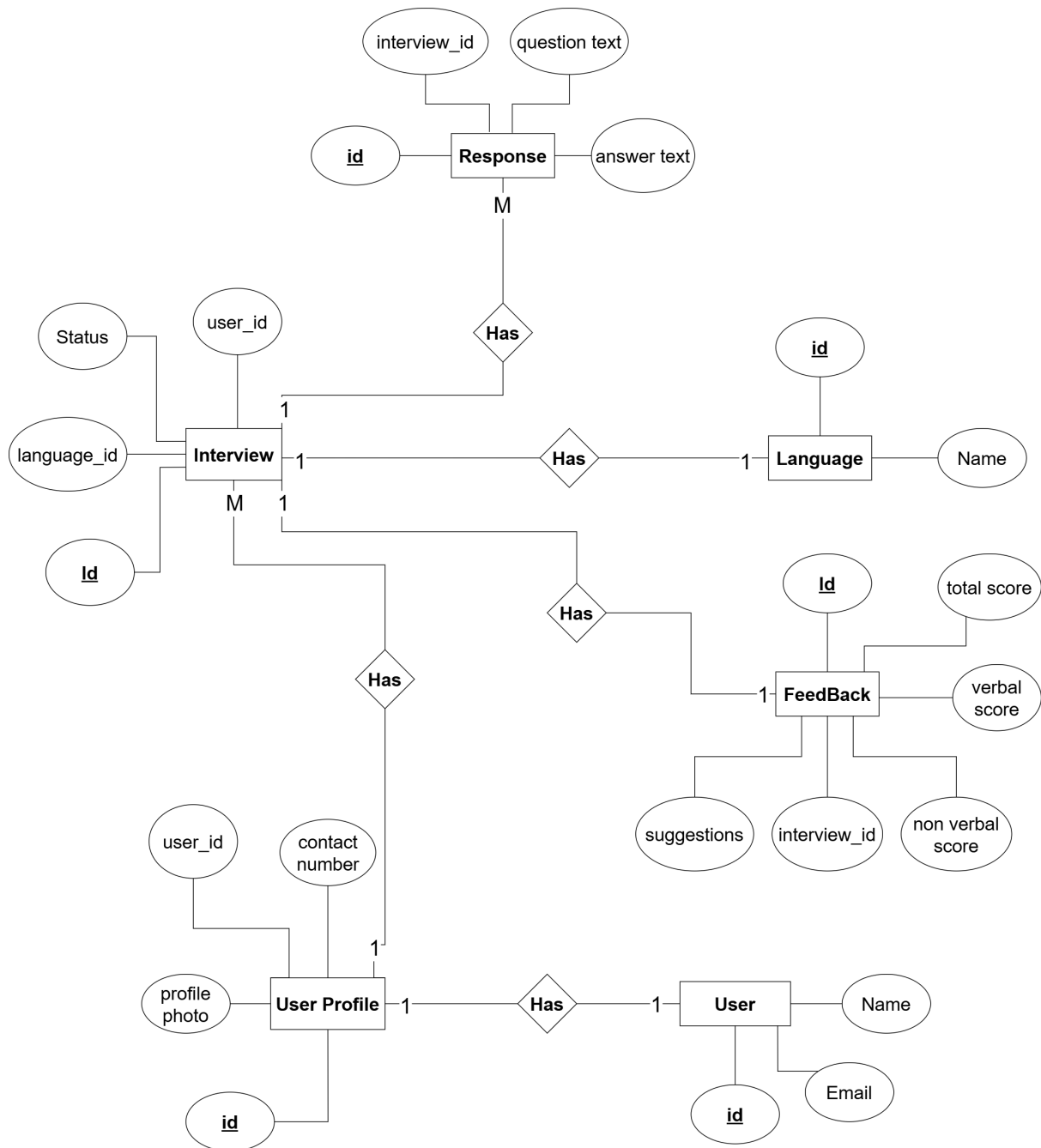


Figure 4.10: Entity Relationship Diagram.

4.9.2 4.9.2 Data Dictionary

The full Data Dictionary of the Nexora system is included in this section. It defines each of the database tables, their attributes, their data types, their relationships, and constraints that make up the data model which is at the core of the system. Each entry has been defined with the intention of clarifying its purpose for the reader for development, maintenance, and future enhancement processes. The tables

described represent all of the major functionality within Nexora, including user management, simulated interviews, response tracking, performance evaluation, and AI-generated feedback and scoring.

Table 4.13: Data Dictionary

Entity	Attribute	Data Type	Relation To	Nullable	Description
User	id	INT (PK)	—	No	Primary key
	Name	VARCHAR(100)	—	No	Full name of the user
	Email	VARCHAR(150)	—	No	User's email address
	Password	VARCHAR(255)	—	No	Hashed password
User Profile	id	INT (PK)	—	No	Primary key
	user_id	INT (FK)	User.	No	Foreign key
	contact number	VARCHAR(20)	—	Yes	User's phone number
	profile photo	VARCHAR(255)	—	Yes	Path/URL to profile picture
Language	id	INT (PK)	—	No	Primary key
	name	VARCHAR(50)	—	No	Language name (e.g., English, Urdu)
Interview	id	INT (PK)	—	No	Unique identifier for interview session
	user_id	INT (FK)	User	No	The candidate giving the interview
	status	VARCHAR(20)	—	No	Status of interview (e.g., ongoing, completed)
	language_id	INT (FK)	Language	No	Language used in interview

Entity	Attribute	Data Type	Relation To	Nullable	Description
Response	id	INT (PK)	—	No	Unique identifier for response
	interview_id	INT (FK)	Interview	No	References the interview
	question text	VARCHAR(255)	—	No	Question asked
	answer text	VARCHAR(255)	—	Yes	Candidate's answer
Feedback Report	id	INT (PK)	—	No	Unique identifier for feedback
	interview_id	INT (FK)	Interview	No	References interview
	total score	DECIMAL(5,2)	—	No	Overall interview score
	verbal score	DECIMAL(5,2)	—	Yes	Score for verbal communication
	non verbal score	DECIMAL(5,2)	—	Yes	Score for non-verbal communication
	suggestions	VARCHAR(255)	—	Yes	Feedback suggestions for improvement

4.10 Risk Analysis

In this section, we highlight some of the potential risks associated with the development, maintenance, and scalability of the Nexora platform.

4.10.1 Technical Risks

Since Nexora will integrate a number of modern artificial intelligence solutions, such as natural language processing, speech recognition, and computer vision, it will face a threat of an accelerated development of technologies. The existing frameworks, APIs, and AI models can be out-of-date or obsolete and require significant upgrades or replacements of the system in the future. Moreover, due to the further development of AI methods and user demands, the platform might need serious revisions in order to maintain its performance, accuracy, and compatibility. Also, the ongoing incorporation of huge data

volumes and the necessity of immediate feedback may expose technical issues related to processing speed, optimisation of models and the ability to scale up the system.

4.10.2 Business Risks

Although at present the application works fine with shared hosting and other types of virtual services, for future versions, if dedicated servers or cloud hosting become viable options to improve accessibility and scalability, the application may require the financial support for that. Moreover, with Nexora's growth, there may be a need for employing devs and acquiring technical resources to ensure proper maintenance and iterations, and also to expand its functionality. Poor financial and technical management with regards to the longevity and improvement of the system would be a problem. Hence, a deliberate and gradual budgeting plan will be necessary to support the growth and stability of the project.

4.10.3 Privacy and Ethical Risks

Sensitive data, including the audio and video responses of users, is processed during mock interviews to deliver fast feedback by Nexora. Despite the fact that the data is analyzed for short-term purposes and not long-term storage, privacy and ethical issues are still raised. Then, intrusion into the personal domain, as well as the unauthorized collection or dissemination of this information, would not only undermine users' trust but may also raise legal concerns. In addition, algorithmic bias could occur due to an absence of diversity in the collected data so that grading and analyses could be inaccurate. To eliminate these potential threats, Nexora will implement data encryption, safe authentication methods, and abide by relevant data protection. The model and the fairness assessment will be subjected to periodic audits to ensure transparency, accountability, and ethical practices of AI in this process.

4.11 Conclusion

In conclusion, this chapter presents a detailed overview of the project's Software Requirement Specifications (SRS). This document identifies the key components, requirements, and quality attributes necessary for the application to be an effective recruitment tool. Particular focus is given to the system's main features, such as flexibility, and security, which contribute to a reliable and user-friendly experience. The use cases and their descriptions show the system's main functionality and the interactions of the users with the system. Furthermore, the ER diagram and the data dictionary add a representation of data organization and flow, supporting the understanding and consistency of the system's architecture. The technology stack proposed for implementation is discussed in detail, ensuring technical feasibility, and the potential risks from both the technological aspect and business aspect are also discussed.

Chapter 5

High-Level and Low-Level Design

This chapter describes the top and bottom levels of the Nexora system, which in turn displays the overall system design, with an emphasis on the largest parts and their links. Moreover, the chapter includes the architecture and class diagrams, which show how different modules, classes, and data entities relate with one another through arrows and boxes. Together, these designs form the blueprint for implementing a robust, scalable, and maintainable system.

5.1 System Overview

Nexora is an AI-based, web-based application that is designed to transform the paradigm of interview preparation. The platform provides an interactive and cognitively adaptive platform which supports simulated interview encounters with an AI-based interviewer. Using real-time audio and video analytics, Nexora evaluates the performance of the candidates, defines areas that need improvement and produces instant feedback that is aimed to supplement verbal and non-verbal communicative skills. In addition to real-time behavioral assessment, the system also performs content-based evaluation of each candidate's answer. After the user finishes speaking, the transcript of the response is processed through a semantic language model that analyzes clarity, relevance, structure, and completeness. The generated evaluation is stored and later contributes to the candidate's overall performance report. The system is dreaming of providing a smooth, automatic and interactive interview process, thus serving multilingual groups and democratising professional interview training to a global clientele.

5.1.1 AI Interviewer Module

Nexora is primarily made up of the AI Interviewer Module. It mimics the role of a human interviewer and asks questions, interprets the answers of the respondents and provides visual feedback on the spot.

Using the sophisticated Natural Language Processing (NLP) and Computer Vision algorithms, the module measures linguistic tone, gestures, and facial expression through visual indicators and also evaluates the content of candidate's answer using semantic LLM and stores scores and feedback in database after answer completion. The system, therefore, provides real-time analytical information related to the performance of the candidate.

5.1.2 Real-Time Feedback System

The module shall be responsible for tracking and assessing candidate behavior during the interview session. In case the AI system identifies that there are proper pattern of communication and proper posture, a green signal will be displayed. On the other hand, a red alert will be highlighted in case distracting behavior, e.g. moving too much, sitting improperly or eye contact is insufficient. This immediate feedback system helps the candidate to change and improve performance during the interview as opposed to looking back.

5.1.3 Multilingual Support System

Before the interview begins, the candidate has the option to choose which language of the AI interviewer they want to converse with, for example, either English or Urdu. This choice then signifies the language of the questions being asked. The candidate may choose to respond in the chosen language or mix languages, both of which will naturally align with their usual speech patterns or code-switching.

The spoken responses are able to be captured and transcribed via a speech-to-text pipeline like Whisper that maintains the original language of content. The transcription is then sent to the LLM with a prompt to provide the evaluation and feedback in English. The LLM identifies the semantic content and provides a score based on its clarity, relevance and other parameters, all in English, and provides performance feedback that is structured and warranted. This design allows the system to support multilingual responses while keeping all aspects of evaluation and reporting consistent, clear, and professional, as well as allowing a simple user interface as no UI-level multilingual translation is necessary.

5.1.4 User Interface and Experience (UI/UX)

The system is based on a user-friendly and intuitive interface that is created with the latest web technologies. A candidate is able to easily register, authenticate, give interview preferences and start a session with few clicks. The interface is focused on simplicity, accessibility and responsiveness, thus ensuring a smooth experience with a wide range of devices and browsers.

5.1.5 Security and Privacy Module

Since Nexora does not store video and audio streams, and processes them in real-time, the system is supplied with powerful encryption and secure authentication tools. Furthermore, the system guarantees the protection of user sessions by advanced security mechanisms and handles all sensitive information in accordance with the latest data transfer techniques.

5.1.6 Performance and Scalability

The system is capable of interviewing multiple candidates concurrently. The modular design in conjunction with REST API interactions creates a robust framework which in the future maybe incremented with additional languages or AI models if necessary.

5.2 Design Considerations

This section describes elements that should be addressed by design in Nexora.

5.2.1 Assumptions and Dependencies

The understanding of Nexora's design is based on a number of assumptions and dependencies that guarantee an uninterrupted operation and a stable performance.

5.2.1.1 Related Software or Hardware

Nexora requires the use of a web camera and microphone, so users should have them available because they are the primary means for recording videos and audios during the interview. It is also important that recent web browsers should be enabled for media streaming in real time. This means that browser such as Google Chrome, Microsoft Edge, and Mozilla Firefox are also supported. Besides, in the given system React-based interfaces are used.

5.2.1.2 Operating Systems

The platform is designed in such way so that it can work seamlessly with the major operating systems, such as Windows, macOS, and Linux. The web app, Nexora, doesn't need downloading or installing, and it's accessible to any device having a browser without needing any specific platform.

5.2.1.3 End-User Characteristics

It presumes that any candidate possesses at least the basic skills of using a computer coupled with a dependable internet connection, which is essential for smooth video and audio streaming.

5.2.1.4 Future Functional Extensions

The system is designed organically for additional enhancement possibilities, such as addition of new languages, integration of mobile platforms, and further improvement of AI algorithms. Future extensions may also include advanced behavioral insights for candidates, such as detailed body-language breakdown, emotion progression graphs, or personalized improvement roadmaps generated from past interview sessions.

5.2.2 General Constraints

Nexora development is a technical subject that has several limitations, including design space interruption due to environmental issues. They influence the total design parameters, usability, and even efficiency of the programs.

5.2.2.1 Hardware and Software Environment

Nexora may have its absolute reliance on the user's access to a camera, microphone, and a high-speed internet connection. These components directly affect user interface and the level of AI-powered analysis relatively.

5.2.2.2 End-User Environment

Website development involves an environment known as web browsers, which require the use of updated browser applications like Google Chrome, Microsoft Edge, or Mozilla Firefox to get the best performance and fit.

5.2.2.3 Availability of Resources

Nexora is built on the third-party APIs and cloud-based AI models. This implies that a breakdown of the API service or limits of access, even if temporary, could affect the real-time emotion and speech analysis capabilities.

5.2.2.4 Standards Compliance

Nexora employs cutting-edge development techniques using React.js for the frontend part and Ruby on Rails for the backend part. The design also ensures code clarity, proper use of libraries for AI model and speeches, and the system is also scalable and maintainable.

5.2.2.5 Interoperability Requirements

In Nexora architecture, communication between the front end built on React.js and the back end with Ruby on Rails is possible using RESTful APIs. Due to changes in the data structure, and API endpoints, non-compatibility between the services can be introduced.

5.2.2.6 Data Repository Requirements

The hosting of all data on a secure PostgreSQL database includes consistency and integrity of the database with multiple users at the same time. The same repository also guarantees confidential storage of user data, materials of the interview, and analyzed results generated by AI.

5.2.2.7 Security Requirements

Nexora is a space-time service that streams audio and video data, which means it should support HTTPS encryption alongside authentication. All of the sensitive data cannot be retained temporarily, and everything is processed according to data protection standards.

5.2.2.8 Performance Requirements

With the system being dependent on live data, Nexora will ensure speedy latency and demand interview timing and feedback-to-solution response from the candidate.

5.2.2.9 Network Communication

The persistence of the internet is critical to avoid disruptions to the AI-powered interviews. Doubtful network connectivity could affect the quality of the user experience or take away sessions to be directly fixed.

5.2.2.10 Verification and Validation

Including unit testing, integration testing, and usability testing Nexora will go through a consistent testing process to confirm that the system possesses accuracy, reliability, and effortless scalability prior to launching.

5.2.3 Goals and Guidelines

The architecture of Nexora is shaped in such a way that set of guiding principles which ensure usability, scalability, and system reliability throughout its development.

5.2.3.1 Simplicity (KISS Principle)

Nexora's system architecture lays out the "Keep It Simple, Stupid" principle: System and interface features are designed straightforward and easy to manage, thus future developers will not have major issues in expanding the system's functionality and overcoming complexity.

5.2.3.2 User-Centric Design

The interface is designed as that for accessibility and ease of navigation. Intuitive layouts and responsive feedback mechanisms which will scale the normal user experience to a higher level.

5.2.3.3 Scalability and Performance

A simultaneous user session is expertly controlled in the system. Improved efficiency of back-end and data processing ensures the reliability of performance in any situation with a high load.

5.2.3.4 Modularity and Maintainability

Many subsystems may be independently developed in accordance with the API standard. The responsibility for UX, AI, or data processing may be allocated in this way. Modularity that also allows for removing or optimizing the process.

5.2.3.5 Security and Data Privacy

Nexora implements a multitude of security measures which include user authentication, encrypted data transfer, and access control. The original media files are completely deleted, and only the analysis results are remain.

5.2.3.6 Reusability and Extensibility

The architecture entails component reuse and supports the new functions, e.g. AI model updates, analytics dashboards, and mobile app subsection.

5.2.4 Development Methods

Nexora will implement the Scrum methodology that will ensure an iterative and flexible workflow. The work will be divided into sprints of one week each, where each sprint will have specific objectives and will be time-bound. Project management software such as Trello or Kanban boards will be used, which will help in progress tracker and accountability. Review meetings will be conducted at the end of each sprint to gauge the progress, define the problems, and plan for the improvements. Such an iterative plan tends to foster teamwork, ongoing software improvement, and openness to new conditions as the project progresses using this process, leading to less coding errors and bugs.

5.3 System Architecture

The Nexora Architecture, depicting the relations between the different components embraced in Nexora make provision for a seamless real-time interview process.. It will be created in such a way as to facilitate modularity, scalability, and effective communication between subsystems, including the Frontend, Backend, AI Module, and Database. The subsystems have specialized roles, which they perform in conjunction with each other through standardized APIs in order to achieve the overall goals of the system.

5.3.1 Overview

Nexora is a service-based, modular architecture, where each module is given a specific functional task.

- The frontend, which will be coded in React.js, is the user interface through which a candidate can be engaged in AI-based mock interview simulators.
- Its backend in Ruby on Rails will serve as the central node, which will manage user sessions, routing data, and communication between modules.
- Performs real-time processing of camera and microphone data to detect behavioral cues and provide instant visual feedback (red/green indicators).
- Any user and session information will be stored safely in a PostgreSQL database hence providing consistency and reliability of information.

Since the system is planned to be used in the academic field, the processing of recordings will be carried out in real time only and not stored permanently to minimize storage costs.

5.3.2 Top-Level Responsibilities

On the top-most level, Nexora has the following core functions entrusted to it:

- **User Management:** Candidates registration, authentication and profile management.
- **Interview Monitoring:** Real-time capture and analysis of video/audio input.
- **AI Feedback Generation:** The behavioral cues (e.g. unnecessary hand movements or evidence of distraction) are detected and the visual indicators are shown (red or green). Answers are also evaluated using content based semantic LLM models.
- **Language Support:** Multilingual interview support to increase the accessibility.
- **Data Handling:** Controlled management of user data, log-in history and feedback outcome.

5.3.3 Rationale

The design used in Nexora takes a modular structure, hence improving maintainability, clarity and performance. All subsystems are independent but they communicate through RESTful application programming interfaces, thus enabling a smooth debugging process, update isolation and future scalability. The monolithic architecture was not chosen as this structure would reduce the flexibility and the ability to introduce new AI models or update the user interface easily. The selected architectural paradigm ensures the efficient processing of real-time, maintains a light footprint, and reduces infrastructural costs, which makes it highly scaled, and practically feasible to modern web-based AI systems.

5.3.4 System Architecture Diagram

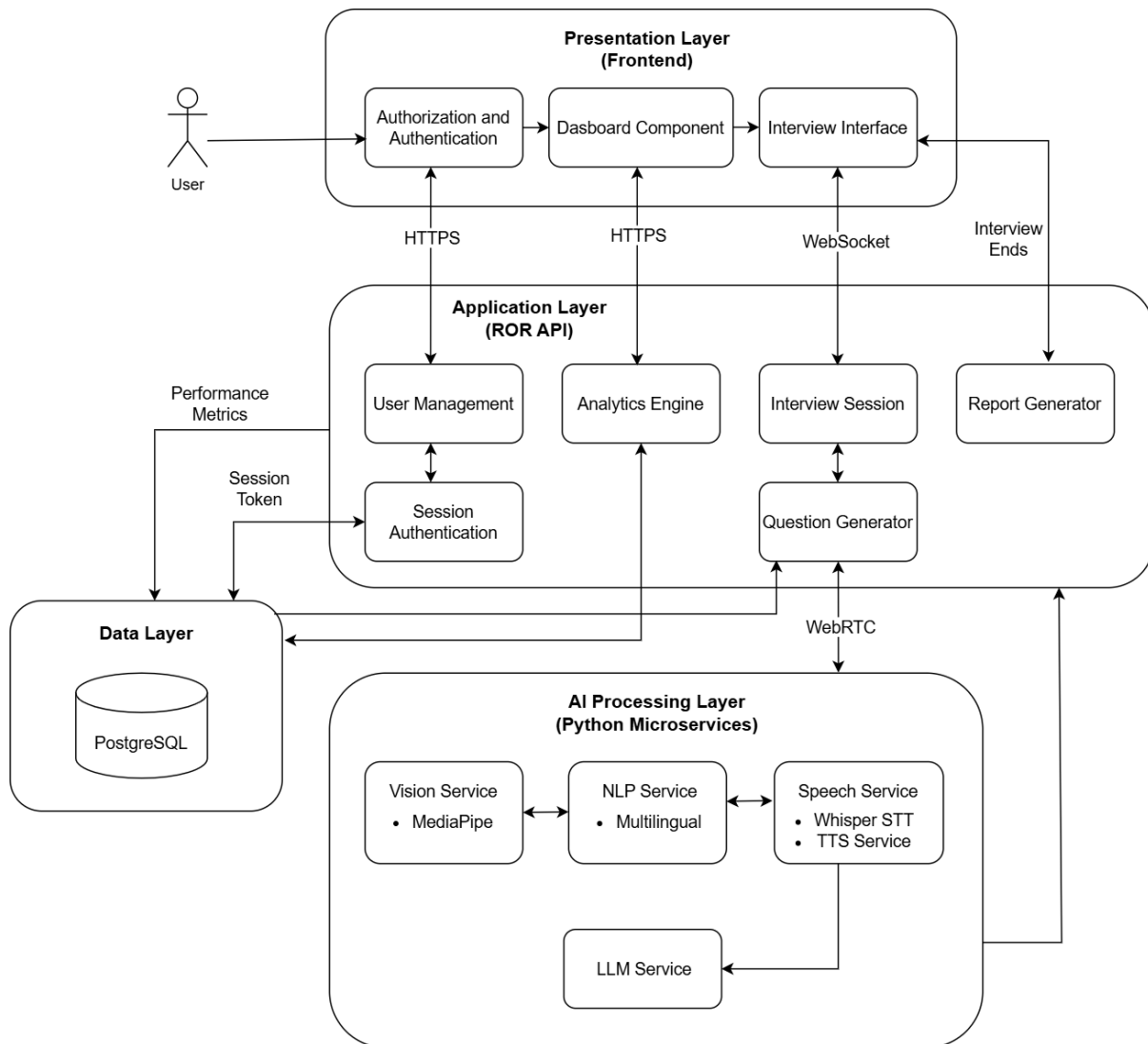


Figure 5.1: Architecture Diagram

5.3.5 Subsystem Architecture

The subsystem architecture of Nexora defines how different subsystems collaborate to deliver the system's overall functionality. This modular approach ensures smooth interaction among components while maintaining scalability, flexibility, and efficient performance.

5.3.5.1 Authentication Subsystem

Authentication Subsystem ensures a secure and smooth access to Nexora platform. It authenticates user identities and administers sessions as well as authorization policies. It also manages password recovery and protects sensitive user data from unauthorized access.

Internal Components:

- User Credentials Management
- Session Handling
- Access Control

Interactions:

- The user enters credentials on the frontend.
- The credentials are checked against the database by the back end.
- Access Control ensures that only authenticated users can access interview functionalities.

5.3.5.2 Interview Management Subsystem

This subsystem regulates the real-time interview process, which will support the unproblematic communication between the front and back end and the AI module. It records the video and audio of the candidate and sends the information to be analyzed and provides real-time feedback. This subsystem also initiates the answer-evaluation process as soon as it determines that the user is done speaking. It then packages this transcript and forwards it to the AI Analysis Subsystem for scoring the semantic content all developed by the language model.

Internal Components:

- Session Control
- Response Segmentation Manager
- Media Stream Handler
- Feedback Display

Interactions:

- The frontend captures audio and video.
- The backend routes this data to the AI module.
- The user receives real-time feedback that is sent back.

5.3.5.3 AI Analysis Subsystem

The AI Analysis Subsystem carries out analysis utilizing NLP and computer vision to provide an evaluation of user behavior and offer real-time analysis. As part of this process, it analyzes facial expressions,

speech patterns, and audio cues to assess attentiveness, confidence, and the quality of communication. Additionally to the multimodal analysis, this sub-system also processes the entire transcript associated with each response once it has been completed and sends it to a semantic LLM for in depth assessment of the content. Once this is done, the associated score data is returned to the backend for storage and integration into the final feedback report.

Internal Components:

- Audio and Visual Processing Unit
- AI Feedback Engine
- Semantic LLM
- Data Communication Interface

Interactions:

- Receives real-time data from the backend.
- Performs analysis using NLP and computer vision.
- Performs content analysis using LLM.
- Sends feedback metrics to the backend for UI update and content based feedback in DB.

5.3.5.4 Data Management Subsystem

This subsystem provides a safe and effective management of structured data which includes user profile, session logs, and AI outputs. Since the system is academic in character, the recordings are never stored permanently to reduce cost and are processed in real-time only.

Internal Components:

- Database Management (PostgreSQL)
- Data Access Layer
- Logging Services

Interactions:

- The backend performs read/write operations on structured data.
- AI analysis results and user development are safely stored.
- Session analysis Logs are kept.

5.4 Architectural Strategies

The architectural strategies that will be used to design and implement Nexora are described below. These strategies define the tools, technologies, and methods that guide the system's development and ensure scalability, maintainability, and real-time performance.

5.4.1 Software Tools and Technologies

The software tools and technologies define the core environment in which Nexora is developed and deployed. These components ensure a balance between performance, scalability, and maintainability while supporting real-time AI-driven interaction.

5.4.1.1 Frontend Framework

React is used to develop a responsive and interactive user interface. It has a component-based architecture that supports reusability, real time UI updates and effective state management during the interview process.

5.4.1.2 Backend Framework

Ruby on Rails provides a robust backend structure to handle authentication, session management, API endpoints, and communication with the AI module. It facilitates fast growth and yet with clean code structure.

5.4.1.3 AI Processing Services

Microservices are written in Python and handle the audio and video files of the candidate in real time. These services perform speech analysis, facial expression tracking, and AI-driven feedback generation using machine learning models. These services also oversee the answer evaluation process by submitting completed transcripts to the LLM and obtaining organized scoring data. The results of the answer evaluation are sent back to the server for inclusion in the final report and ultimately in the flow of the total analysis.

5.4.1.4 Database Management

PostgreSQL is used as the relational database to store the structured data like user profiles, interview records and AI feedback securely. It is scalable, complies with the ACID, and is reliable.

5.4.1.5 Platform Access Requirements

Nexora is a web-based system and not local based. The users can use the site with the help of a modern web browser and a stable internet connection to facilitate real-time communication and streaming.

5.4.2 Hardware

This section outlines the hardware resources necessary for both end users and developers to effectively run and test Nexora. Appropriate hardware ensures smooth real-time video and audio processing and reliable system performance during interviews.

5.4.2.1 End-User Hardware

Nexora will be available through laptops or desktop computers that have webcams and microphones. Such tools are mandatory when it comes to capturing video and audio in the course of the interview session.

5.4.2.2 Development and Testing Hardware

Medium-tier computing hardware should be used in the development and testing stages, so that real-time data streaming, artificial intelligence processing, and performance assessment are properly handled.

5.4.3 Future Plans

This section describes the proposed enhancements and future directions for Nexora. These planned developments aim to extend system functionality, improve scalability, and enhance the overall user experience in upcoming versions.

5.4.3.1 Integration with Recruitment Platforms

Subsequent versions will incorporate Nexora with professional recruitment or HR management systems. This kind of integration will enable institutions and organizations to perform mass automated interviews and performance analytics archiving to select.

5.4.3.2 Cloud Scalability and Analytics Dashboard

To facilitate institutional deployment, Nexora will integrate scalable deployment to the cloud and supply a detailed analytics dashboard of aggregated performance measurements and comparative data. Such proposed improvements will guarantee that Nexora can be extended and placed in a position that it can become a full-fledged AI-driven interview and assessment system.

These future plans ensure Nexora remains extensible and ready to evolve into a comprehensive AI-driven interview and assessment ecosystem.

5.5 Domain Model/Class Diagram

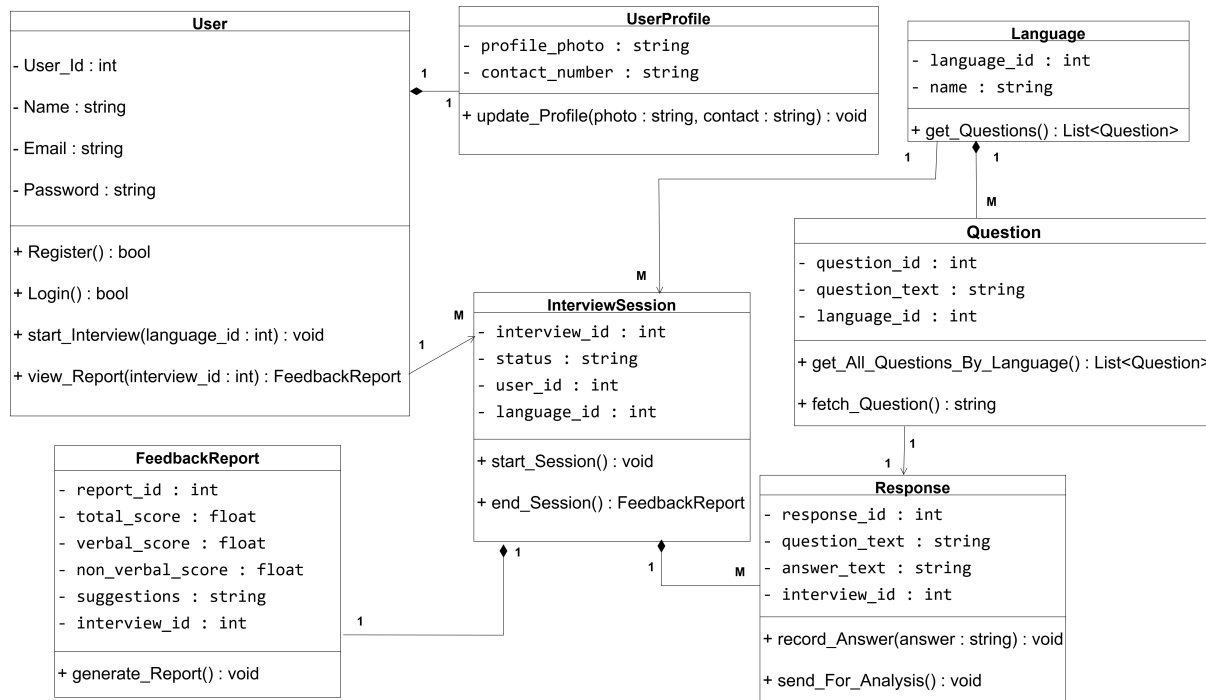


Figure 5.2: Class Diagram

5.6 Policies and Tactics

To ensure the robustness and efficiency of the final application, the project employs various industry standards and policies. The following are some of the policies that we will follow during the software life cycle.

5.6.1 Conventions and Coding Guidelines

React.js, Ruby on Rails and Python have defined sets of coding standards that will be followed to ensure that all modules remain readable and consistent. Nomenclature of descriptive variables and functions will be used with proper indentation and extensive commenting. The components will be properly documented in-house to support teamwork and maintenance. Code reviews will be done frequently to maintain quality and best practice.

5.6.2 Coding Environment

VS Code will be the main development environment, as it has a large number of React.js, Rails, and Python extensions. It has in-built debugging, linting and formatting facilities thus aiding in the maintenance of clean and error-free code. The terminal integration included in VS Code allows developers to run Rails commands, handle dependencies and test modules efficiently.

5.6.3 System Testing

The approach will involve a hybrid testing, which is a combination of white-box and black-box approaches to ensure reliability and accuracy of the system. Unit testing will be used to test that individual components, frontend, backend, and AI services, work, and integration testing will test that they communicate with one another. User acceptance testing (UAT) will be done to ensure that the system meets the intended requirement and provides a smooth user experience.

5.6.4 Maintenance of Software

After the development, the project will have a periodic dependency, framework, and model component updates. The further work will be done on continuous improvement to solve minor bugs, improve performance, and refine AI models based on stakeholder feedback. Code audits will also become periodical to ensure long-term stability and sustainability of the system.

5.7 Conclusion

Nexora is a web-based application that is designed to give the intelligent and interactive mock interview experience. Its design ensures modularity, scalability and ease of use, thus making it easy to operate and easier to enhance in the future. The carefully designed class and architecture diagrams also enhance maintainability and reusability, hence forming a sound basis of an efficient and intelligent interview preparation platform.

Chapter 6

Implementation and Test Cases

This chapter focuses on the development of the application side, where we have built the main structure of our prototype and included some of the key functionalities that form the pillars of our entire system. At this stage, the goal is not to develop the final product but to build the minimal and essential components, the necessary workflows, and the interfaces for the full product development to enter the next stage. The upcoming sections will give a brief overview of the prototypes we have created until now, the technologies that we have harnessed, and the procedural logic that has been implemented during this phase.

6.1 Implementation

6.1.1 Authentication Module

Primarily, the authentication module protects the application against all the unauthorized access. JWT-integrated login, sign-up, and logout were accomplished. The backend of Rails verifies the sign information and sends that over as a signed token to React as front end. This token is automatically stored in the client application and is used to enforce route protection within the React frontend. Any user who is not authenticated is redirected to the sign-in page before accessing the dashboard or interview interface.

6.1.2 User Interface and Dashboard

The adoption of the Tailwind CSS framework resulted in a new user interface that includes trending and adaptive layouts for the landing page, authentication screen, and dashboard. To increase the usability of the platform for the end user, status indicators were put in place; these include loading states, error messages, and success responses. The dashboard can only be accessed after successful authentication,

and access is granted only when a valid token is detected.

6.1.3 Real-Time Interview Prototype

The MediaStream API of the browser brought life into the foundations of a real-time interview tool Nexora. The system is able to use the user's webcam, capture video frames, and send them to the back-end API, which now serves as a placeholder. A green/red indicator emulating the traditional feedback mechanism is now available to stand in for the future AI-powered version to perform the analysis in such an environment.

6.1.4 Backend Configuration and Prototype APIs

We applied the required CORS settings, and also has implemented the mailer development support, and defined the RESTful API routes for user authentication and interview data streaming. On the other side of the client side, there was a basic setup process that involved installing build tools and dependencies that are required to have the client-server communication.

6.1.5 Continuous Integration Pipeline

Initially, a CI/CD pipeline was created without requiring deep coding knowledge, with the purpose of reducing manual effort during development. With the every push, the pipeline automatically starts installing packages, formatting code with Prettier, running tests, and many more tasks are performed. As a result, code changes remain continuously valid and consistent with the application's requirements and the development standards defined for the project. Moreover, the pipeline addresses not only building and deployment but also creating artifacts which are production ready. This allows for real-time continuous integration and resolves the factor of blending issues as the development progresses.

6.2 Conclusion

Now, we have reached the stage of our system's primary prototyping, and testing for proving the concept of this idea is already on the course. All the essential building blocks, such as authentication, user interface, real-time interview prototype, backend setup, and CI pipeline, are established in order to build this project on top of a rock-solid foundation. And although AI-driven final analysis is something we are still working on, including all major components, workflows, and integrations it needs for future improvement, the current building blocks are already in place. This provides the flexibility to incorporate additional functionalities and enhancements that will be developed in the next phase of the project.

Chapter 7

Conclusion and Future Work

This chapter concludes by reiterating the work done and prototype developed so far and also highlights the plans we intend to work on in the future.

7.1 Conclusion

This chapter summarizes the work completed during FYP-1 and presents reflections on the development of Nexora, a real-time interview readiness system that assists users during practice interviews by detecting non-verbal and verbal behaviors such as excessive hand movements, poor body language, or unclear speech etc and providing instant audio-visual feedback through a simple color-based visual cue.

In this first phase of the project, we focused on laying the technical and structural foundation for the system. This included defining the functional requirements, documenting the complete system flow, identifying suitable technologies for real-time processing, and designing the architecture that will support Nexora's interactive behavior monitoring features. The work completed in FYP-1 aligns well with original objectives. We had successfully analyzed the available solutions, identified some key gaps like real-time guidance and support for non-verbal cues, and clarified how Nexora fills those shortcomings. Also, the scope of the system was clearly outlined, and all design-related tasks like identification of the modules, creation of workflows, and architectural planning were completed as planned. Since FYP-1 focuses on the plan and not on the implementation, integration of the models in real-time and development of an interface were kept intentionally out and shifted to the next phase. In general, all the objectives of this phase were fully realized. In the course of this phase, a number of technical and planning challenges were encountered. First was determining the most efficient approach for real-time gesture and behavior detection in a web environment where performance and latency are crucial. The second one was the choice of techniques to handle multilingual feedback generation while maintaining the lightweight and

browser-friendly nature of the system. Identifying these complexities at this early stage has helped refine the system design for smoother implementation in FYP-2. Full model integration, live visual cue engine, multilingual support, and responsive UI were not implemented yet because they fall into the development scope of the next phase. In the end, FYP-1 delivered the conceptual and structural work of Nexora. The project now has a defined architecture, a set of well-defined functionalities, and a solid execution plan. Through this phase, we developed a better understanding of the technical challenges that may be presented to us, the performance considerations needed for real-time systems, and how such hurdles may be surmounted. Full coverage of the scope of FYP-1 was achieved, and the system is now ready for the development stage where its practical feasibility shall be demonstrated in the form of an operational prototype. Nexora stands a good chance of revolutionizing interview preparation through instant, actionable insights which would make candidates more confident and aware of their talking habits

7.2 Future Work

Besides the core prototype, there are several possible improvements that could enhance further the effectiveness of Nexora. but are not part of the assured FYP scope. A mobile version of the system could enable users to practice interviews on the go. Advanced posture analysis and full-body movement tracking could Provide deeper behavioral insights, and the system could be further extended to support other languages. or incorporate adaptive learning mechanisms that customize the interview questions based on user performance. In the long term, Nexora could evolve into a full interview coaching platform with even more analytics and professional recruitment integration. Such improvements would increase the scope and effectiveness of the application following the development of the core system.

7.3 Planned Work for FYP-2

Based on the outcomes of FYP-1, the following structured plan outlines the development activities to be completed in the next phase:

1. Finalize the frontend, backend, and AI module architecture.
2. Implement real-time webcam and microphone connectivity within the web application.
3. Developing gesture recognition, hand movement tracking, and facial activity monitoring algorithms for use in browsers.
4. Integrate the color coded feedback mechanism that responds instantly to distracting behavior.

5. Implement the candidate answer content evaluation module using LLM to generate semantic scores, feedback, and answer-quality analysis.
6. Implement multilingual NLP modules that allow interviews to be conducted in different languages.
7. Develop the complete user interface, including the practice environment and feedback display.
8. Implement analytics dashboards and performance trends to provide actionable insights on user behavior.
9. Optimize processing performance to ensure low latency, real time audio visual analysis.
10. Conduct comprehensive system testing, debugging, and refinement based on user trials.
11. Prepare documentation and deployment plans for the near final prototype.

This plan ensures that FYP 2 will transform Nexora from a conceptual framework into a functional prototype capable of demonstrating its full capability as an intelligent interview preparation tool.

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